Computer Network (CT 702)
BCT IV/I

Binod Sapkota
Asst. Professor
Department of Electronics and Computer Engineering
TU, IOE, Thapathali Campus, Kathmandu

2. Physical Layer (5 hours)

2.1 Network monitoring: Bandwidth, throughput, delay, round-trip-time

2.2 Multiplexing - FDM, TDM, WDM

2.3 Switching - circuit switching, packet switching, datagram and virtual circuit switching

2.4 Transmission media - Guided (twisted pair, coaxial, optical fiber) and unguided media (radio waves, microwaves, infrared)

2.5 Ethernet cable standards (twisted pair and optical fiber)

2.6 Data encoding: Manchester, NRZ-I, MLT3, 4B/5B

Features of Physical layer

- It is the lowest layer of the OSI Model.
- It activates, maintains and deactivates the physical connection.
- It is responsible for transmission and reception of the unstructured raw data over network.
- Voltages and data rates needed for transmission is defined in the physical layer.
- It converts the digital/analog bits into electrical signal or optical signals.
- Data encoding is also done in this layer.
- Representation of Bits: Data in this layer consists of stream of bits. The bits must be encoded into signals for transmission. It defines the type of encoding i.e. how 0's and 1's are changed to signal.
- Data Rate: This layer defines the rate of transmission which is the number of bits per second.

Functions of Physical layer…

- Synchronization: It deals with the synchronization of the transmitter and receiver. The sender and receiver are synchronized at bit level.
- Encoding and Signaling: The physical layer is responsible for various encoding and signaling functions that transform the data from bits that reside within a computer or other device into signals that can be sent over the network.
- Line Configuration: This layer connects devices with the medium: Point to Point configuration and Multipoint configuration.
- Topologies and Physical Network Design: Devices must be connected using the following topologies: Mesh, Star, Ring and Bus.
- Interface: The physical layer defines the transmission interface between devices and transmission medium.

Functions of Physical layer…

- Definition of Hardware Specifications: The details of operation of cables, connectors, wireless radio transceivers, network interface
- Cards and other hardware devices are generally a function of the physical layer (although also partially the data link layer…).
- Data Transmission and Reception Modes: Physical Layer defines the direction of transmission between two devices: Simplex, Half Duplex, Full Duplex.
- Deals with baseband and broadband transmission

2.1 Network monitoring: delay, latency, throughput

- Bandwidth:
  - determines how fast data can be transferred over time.
  - is the maximum amount of data that can be transferred per second from one point to another in a given time period.
  - the number of packets that can be transferred throughout the network.
- For Digital Devices
  - Channel Capacity
  - Bandwidth is expressed in bits per second (bps)
- For Analog Devices
  - refers to the range of frequencies in a composite signal or the range of frequencies that a channel can pass.
  - Bandwidth is Expressed in cycle per sec OR Hz

Throughput

- is the average rate of successful message that, a communication channel can delivery over a communication period.
- the number of packets that are processed within a specific period of time.
- measured unit is bits per second(bps).
- is the number of messages successfully delivered per unit time.
- is controlled by available bandwidth, as well as the available signal-to noise ratio and hardware limitations.

Throughput …

- throughput is the term used to refer to the quantity of data being sent that a system can process within a specific time period.
- **Throughput is a good way to measure the performance of the network connection** because it tells you how many messages are arriving at their destination successfully.
- If the majority of messages are delivered successfully then throughput will be considered high.
- In contrast, a low rate of successful delivery will result in lower throughput.
- The lower the throughput is, the worse the network is performing.

Delays

- Delay is the time required for a signal to traverse the network
- how long it takes for a bit of data to travel across the network from one node (host or router) or endpoint to another.
- As a packet travels from one node to the subsequent node along the path, the packet suffers from several different types of delays at each node along the path.
  - (i) Nodal processing delay /Processing Delay,
  - (ii) Queuing delay,
  - (iii) Transmission delay and
  - (iv) Propagation delay.

Nodal processing delay /Processing Delay

- The time required to examine the packet's header and determine where to direct the packet is part of the processing delay.
- The processing delay can also include other factors, such as the time needed to check for bit-level errors in the packet that occurred in transmitting the packet's bits from the upstream router to router
- *d*proc: nodal processing – processing in a node
  - check bit errors
  - determine output link
  - typically < msec

Queuing delay

- After this nodal processing, the router directs the packet to the queue that precedes the link to router B.
- At the queue, the packet experiences a queuing delay as it waits to be transmitted onto the link.
- The queuing delay of a specific packet will depend on the number of other, earlier-arriving packets that are queued and waiting for transmission across the link; the delay of a given packet can vary significantly from packet to packet.
- If the queue is empty and no other packet is currently being transmitted, then our packet's queuing delay is zero
- *d*queue: queueing delay - packets *queue* in router buffers
    - time waiting at output link for transmission
    - depends on congestion level of router

Transmission Delay

- The amount of time required to transmit all of the packet's bits into the link.
- *d*trans: transmission delay:
  - *L*: packet length (bits)
  - *R*: link *bandwidth (bps)*
  - *dtrans = L/R*

Propagation Delay

- The time required to send packet from one router to another router. The bit propagates at the propagation speed of the link.
- The propagation speed depends on the physical medium of the link and the distance from source to destination.
- *d*prop: propagation delay:
  - *d*: length of physical link
  - *s*: propagation speed in medium (~2×10^8 m/sec)
  - *d*prop = *d*/*s*
- Total Delay /Latency ?
  - **Total Delay dtotal = dproc + dqueue + dtrans + dprop**
  - If there is N node and all the delays are same then the total delay is
    - **Total Delay dtotal = N (dproc + dqueue + dtrans + dprop)**

Latency

- Measures of time delay experienced in a system.
- **Latency indicates how long it takes for packets to reach their destination from the source**.
- **Latency** is used to measure **how quickly these conversations take place**. The more latency there is, the longer these conversations take to hold.
- The level of latency determines the maximum throughput of a conversation. **Throughput is how much data can be transmitted within a conversation**.
- Generally, this is **measured as a round-trip** but it is often measured as a **one-way** journey as well.
- Total latency: One-way latency from source to destination plus the one-way latency from the destination back to the source.
- *Network latency is the delay that is introduced by the network*

What are typical values for latency?

- Typical, approximate, values for latency that you might experience include:
- 800ms for satellite
- 120ms for 3G cellular data
- 60ms for 4G cellular data which is often used for 4G WAN and internet connections
- 20ms for an Multiprotocol Label Switching (**MPLS**) network such as BT IP Connect, when using Class of Service to priorities traffic
- 10ms for a modern Carrier Ethernet network such as BT Ethernet Connect or BT Wholesale Ethernet in the UK

2.2 Transmission media: Twisted pair, Coaxial, Fiber optic, Line-of-site, Satellite

- Cables are commonly used to carry communication signals within **Local Area Networks (LAN)**
- Three common types of cable media
  - twisted-pair cable,
  - coaxial cable, and
  - fiber-optic cable

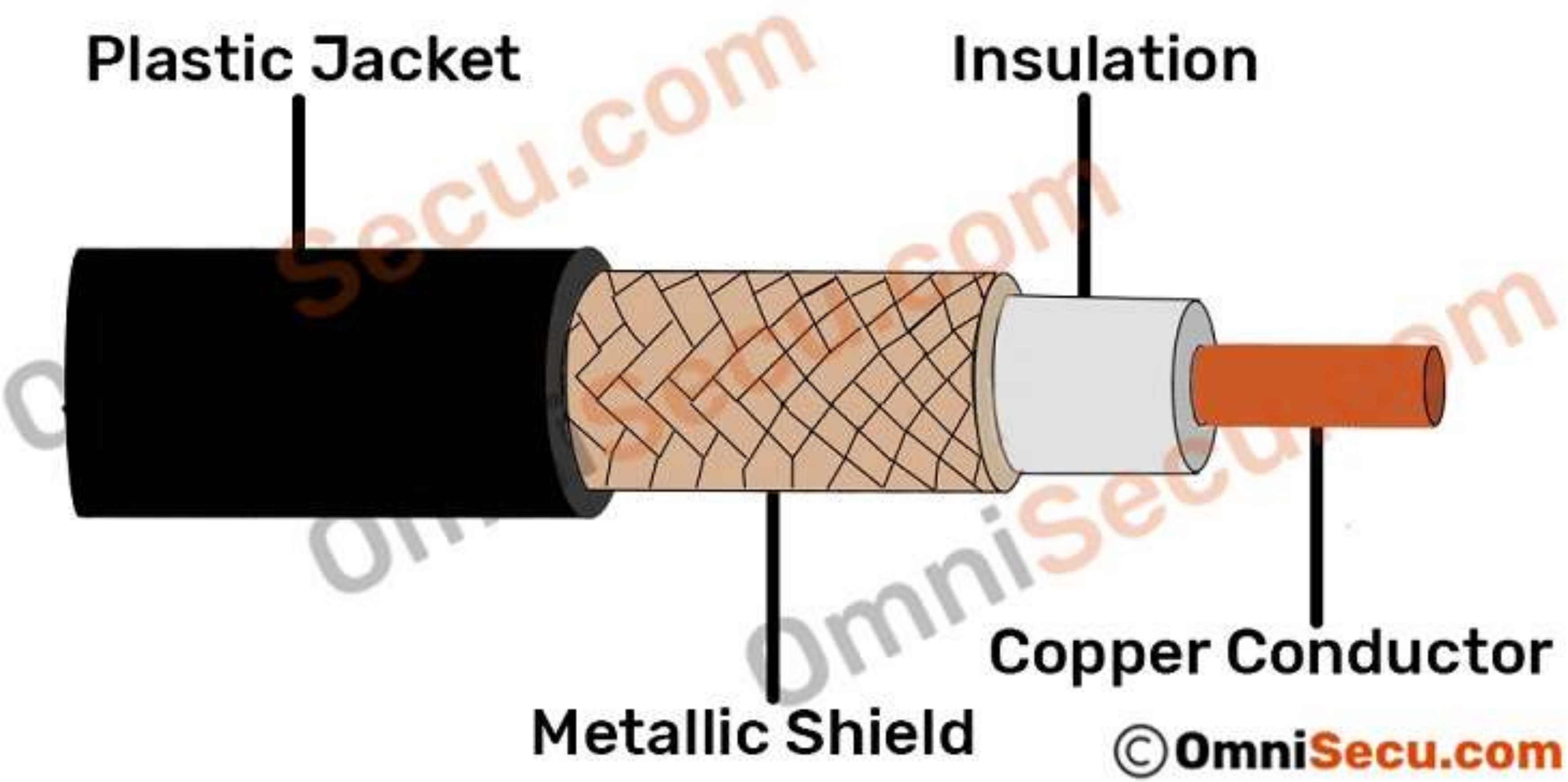
Transmission Medium

- Anything that can carry information from a source to a destination
- Classes (Types)
  - Guided Transmission Media
  - Unguided Transmission Media
- Guided Transmission Media
  - provide a physical path along which the signals are propagated.
    - Twisted pair
    - Coaxial
    - Fiber

Transmission Medium…

- Unguided Transmission Media
  - employ an antenna for transmitting through air or vacuum
- Bluetooth
- Wi-Fi /WLAN (Wireless LAN)
- Infrared
- Satellite Communication
  - Microwaves
  - Radio Waves

General structure of coaxial cable



Two types of coaxial cabling

- **ThinNet**
  - ThinNet is a flexible coaxial cable about ¼ inch thick.
  - ThinNet is used for short-distance.
  - ThinNet connects directly to a workstation's network adapter card using a British Naval Connector (BNC).
  - The maximum length of thinnet is 185 to 200 meters.
- **ThickNet**
  - ThickNet coaxial cable is thicker cable than ThinNet.
  - ThickNet cable is about ½ inch thick and can support data transfer over longer distances than ThinNet.
  - ThickNet has a maximum supported cable length of 500 meters and usually is used as a backbone to connect several smaller ThinNet-based networks.

Two Ethernet media

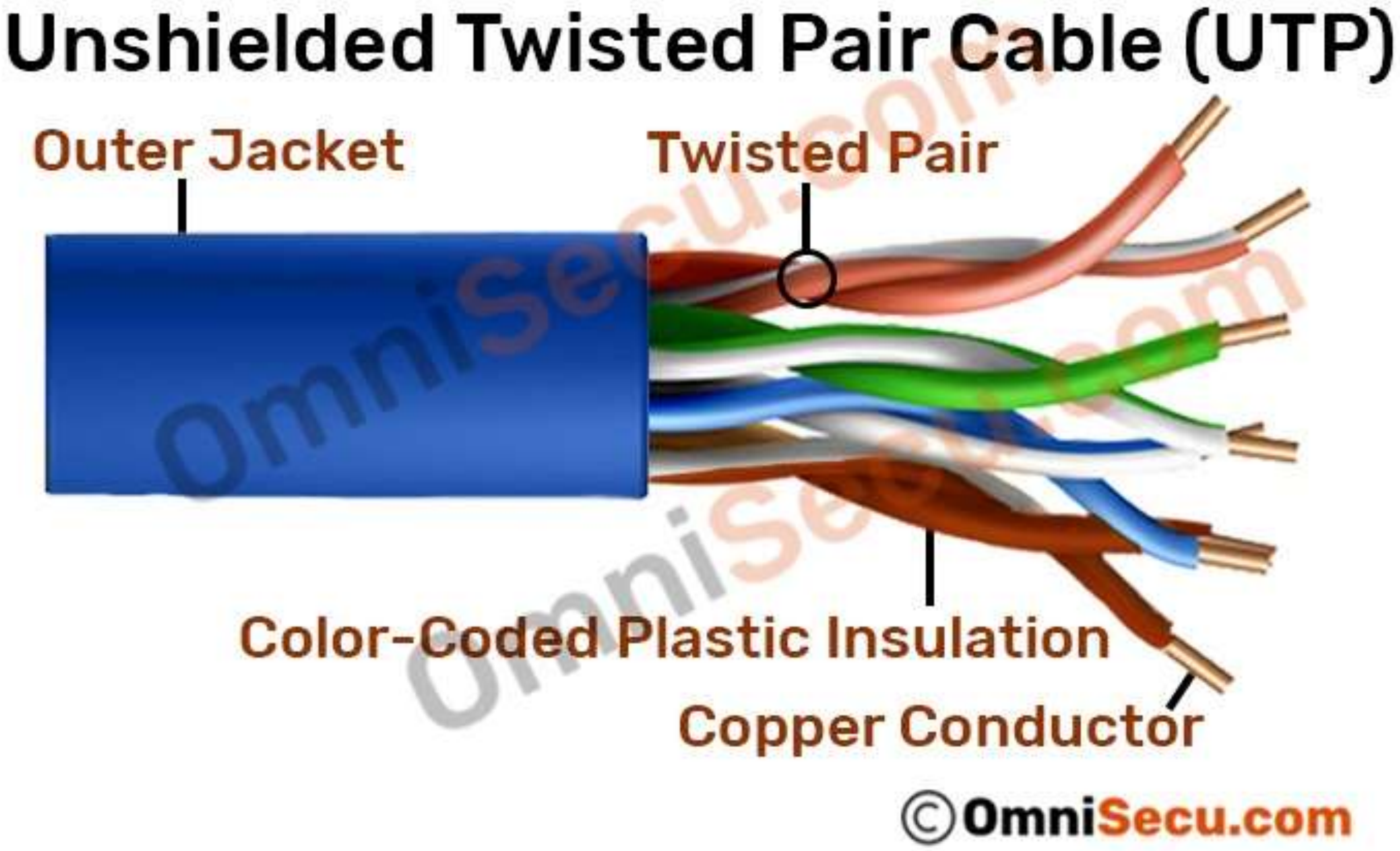
- Two Ethernet media standards defined for coaxial cable-based Ethernet.
- **10Base2**
- 10Base2 has a bandwidth speed of 10 Mbps, to a maximum distance of 200 meters.
- Base denotes **baseband type of signal**.
- Coaxial cable used for 10Base2 Ethernet media standard is ThinNet.
- **10Base5**
- 10Base5 has a bandwidth speed of 10 Mbps, to a maximum distance of 500 meters.
- Base denotes **baseband type of signal**. Coaxial cable used for 10Base5 Ethernet media standard is ThickNet.
- The bandwidth available for both 10Base2 (Thinnet Ethernet) and 10Base5 (Thicknet Ethernet) were 10 Mbps (Megabits per second).

Twisted Pair cable

- Most common type of Cable used to wire Local Area Networks (LAN) in these days is Twisted Pair cable.
- A pair of wires forms a circuit that can transmit data.
- The pairs are twisted to provide protection against crosstalk.
  - Crosstalk is the undesired signal noise generated by the Electro-Magnetic fields of the adjacent wires.
- When a wire is carrying a current, the current creates a magnetic field around the wire. This field can interfere with signals on nearby wires. To eliminate this, pairs of wires carry signals in opposite directions, so that the two magnetic fields also occur in opposite directions and cancel each other out. This process is known as cancellation.

Unshielded Twisted Pair (UTP)

- UTP, the most common networking media consists of four pairs of thin, copper wires covered in color-coded plastic insulation that are twisted together.
  - The wire pairs are then covered with a plastic outer jacket.
- UTP cables are of small diameter and it doesn’t need grounding.
- Since there is no shielding for UTP cabling, it relies only on the cancellation to avoid noise.
- Colors used for Twisted Pair wires are Orange, Orange-White, Blue, Blue-White, Green, Green-White, Brown and Brown-White.


Unshielded Twisted Pair Cable (UTP)
Outer Jacket
Twisted Pair
Color-Coded Plastic Insulation
Copper Conductor
©OmniSecu.com

UTP…

- The connector used on a UTP cable is called as RJ-45 (Registered Jack 45) connector.
- Eight color-coded wires inside Twisted-Pair cable is attached to eight pins in a RJ45 jack as shown below. Each wire in the Twisted Pair cable is crimped into 8 pins in the RJ45 jack.

UTP categories and corresponding transfer rate

UTP Category	Purpose	Frequency (BW)	Transfer Rate
Category 1	Voice Only		
Category 2	Data	4 MHz	4 Mbps
Category 3	Data	16 MHz	10 Mbps
Category 4	Data	20 MHz	16 Mbps
Category 5	Data	100 MHz	100 Mbps
Category 5e	Data	100 MHz	1 Gbps
Category 6	Data	250 MHz	Upto 10 Gbps
Category 6a	Data	500 MHz	Upto 10 Gbps
Category 7	Data (STP version of 6a)	600 MHz	Upto 10 Gbps
Category 8	Data (STP)	2 GHz (2000 MHz)	25 to 40 Gbps

Shielded Twisted Pair (STP)

- Electromagnetic noise penetration is prevented by metal casing. Shielding also eliminates crosstalk
- It has same attenuation as unshielded twisted pair.
- *Advantages :*
  - Easy to install
  - Performance is adequate
  - Can be used for Analog or Digital transmission
  - Increases the signaling rate, faster than unshielded and coaxial cable.
  - Higher capacity than unshielded twisted pair
  - Eliminates crosstalk
- *Disadvantages :*
  - Difficult to manufacture
  - Heavy
  - more expensive than coaxial and unshielded twisted pair


Shielded Twisted Pair Cable (STP)
Outer Jacket
Overall Shield
Twisted Pair
Pair Shields
Drain Wire
Color-Coded Plastic Insulation
Copper Conductor
©OmniSecu.com

Differences between STP and UTP twisted pair cables

Shielded Twisted Pair	Unshielded Twisted Pair
STP - Shielded Twisted Pair cable.	UTP - Unshielded Twisted Pair cable.
STP is costlier in price than UTP.	UTP is cheap in price compared with STP.
STP require grounding of cable.	UTP requires no grounding of cable.
STP reduces electromagnetic interference more than UTP	Electromagnetic interference is more in UTP
Low crosstalk in STP	Crosstalk in UTP is more than STP
STP is fast compared with UTP	UTP is slow compared to STP

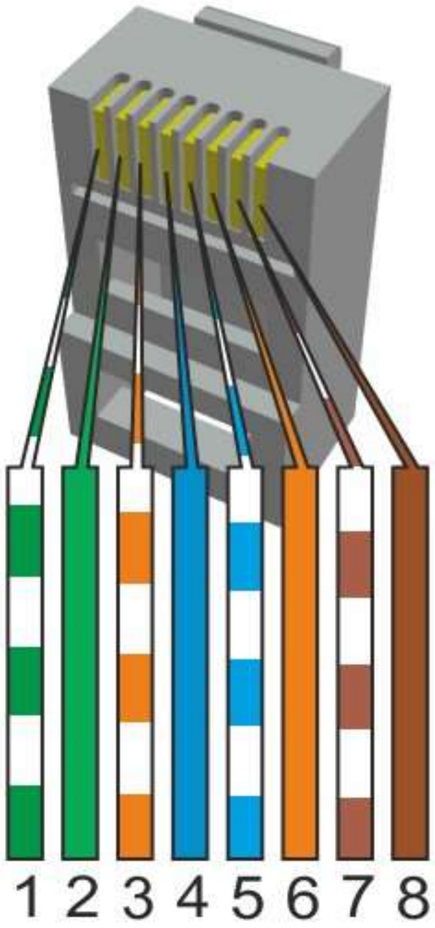
RJ-45 Connector Pinouts -Type A

- T568A pin: $G_wGO_wBB_wOB_wB$ or GOBOB
- RJ-45 odd pin : color with white stripe
- Even Pin: solid Color
- Green and orange colors are used to transmit and receive
  - Green pair (1 & 2): transmits
  - Orange pairs(3& 6): receives
- Blue and brown are not used in 10Base-T and 100Base-T (fast ethernet) networks
- Cross over (1-3, 2-6) and straight through cable

https://www.youtube.com/watch?v=ZrG4yqmRJUw

RJ-45 Connector Pinouts -Type B

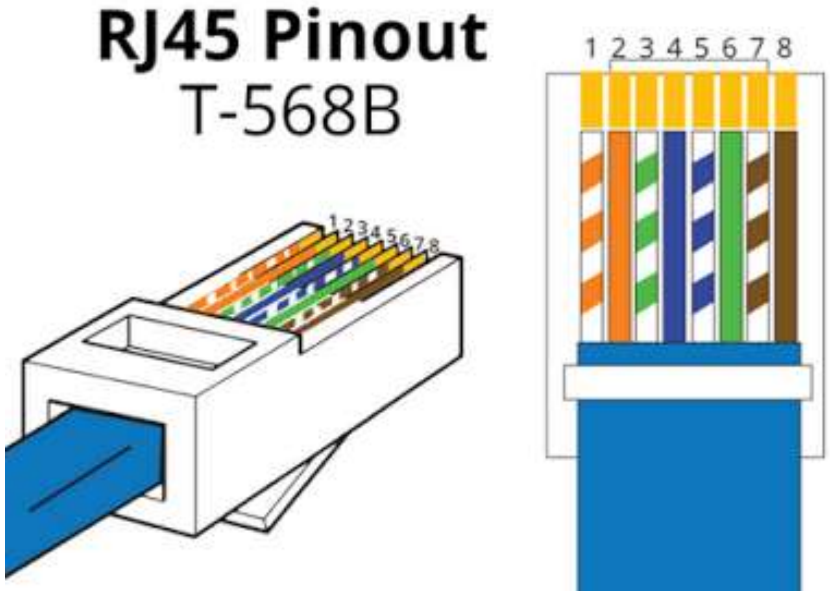
- Pin assignment order: GO swap : green and orange are swapped
- RJ-45 odd pin : color with white stripe
- Even Pin: solid Color
- Blue and brown colors are in the same position
- Green and orange colors are used to transmit and receive
  - orange pair (1 & 2): transmits
  - Green pairs(3& 6): receives
- Blue and brown are not used in 10Base-T and 100Base-T (fast ethernet) networks
- https://www.youtube.com/watch?v=ruH3LofMr4M

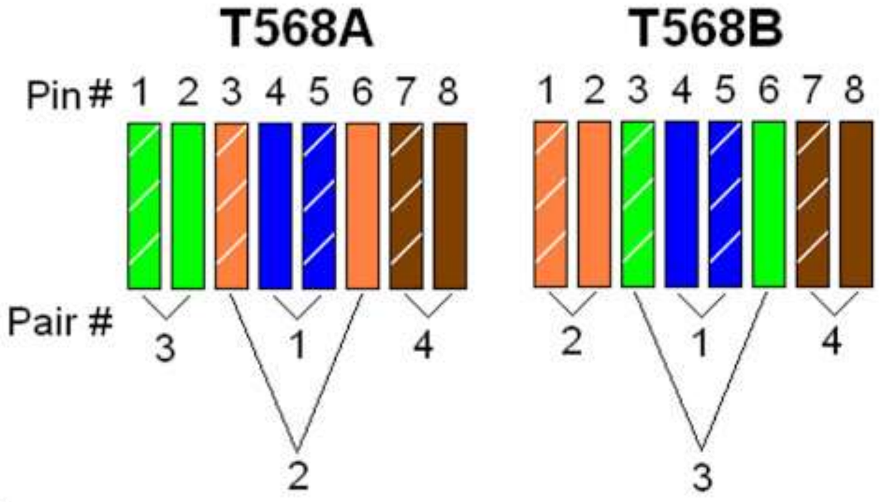


T-568A

Pin	Description	10base-T	100Base-T	1000Base-T
1	Transmit Data+ or BiDirectional	TX+	TX+	BI_DA+
2	Transmit Data- or BiDirectional	TX-	TX-	BI_DA-
3	Receive Data+ or BiDirectional	RX+	RX+	BI_DB+
4	Not connected or BiDirectional	n/c	n/c	BI_DC+
5	Not connected or BiDirectional	n/c	n/c	BI_DC-
6	Receive Data- or BiDirectional	RX-	RX-	BI_DB-
7	Not connected or BiDirectional	n/c	n/c	BI_DD+
8	Not connected or BiDirectional	n/c	n/c	BI_DD-

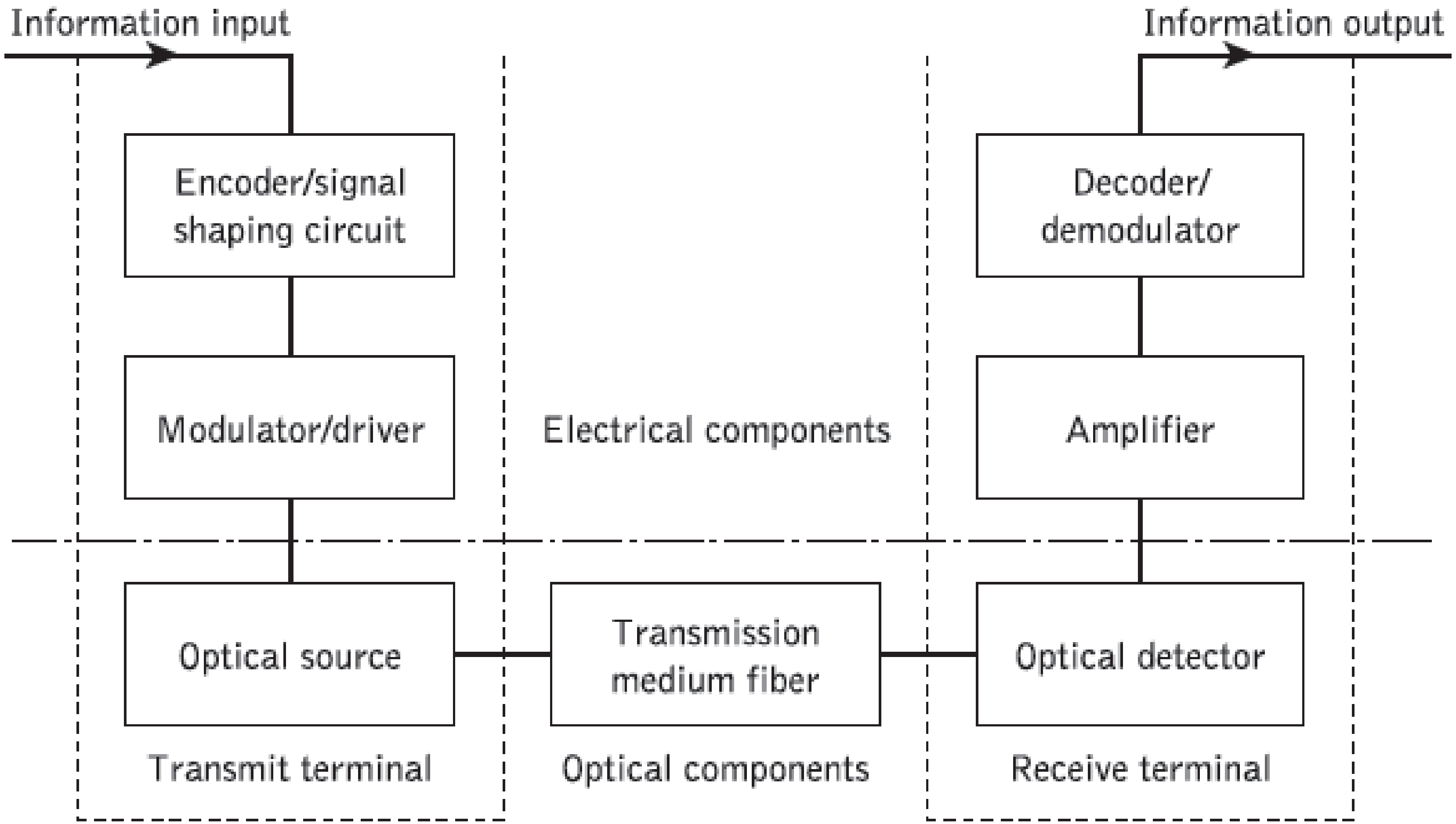
PinoutsGuide.com



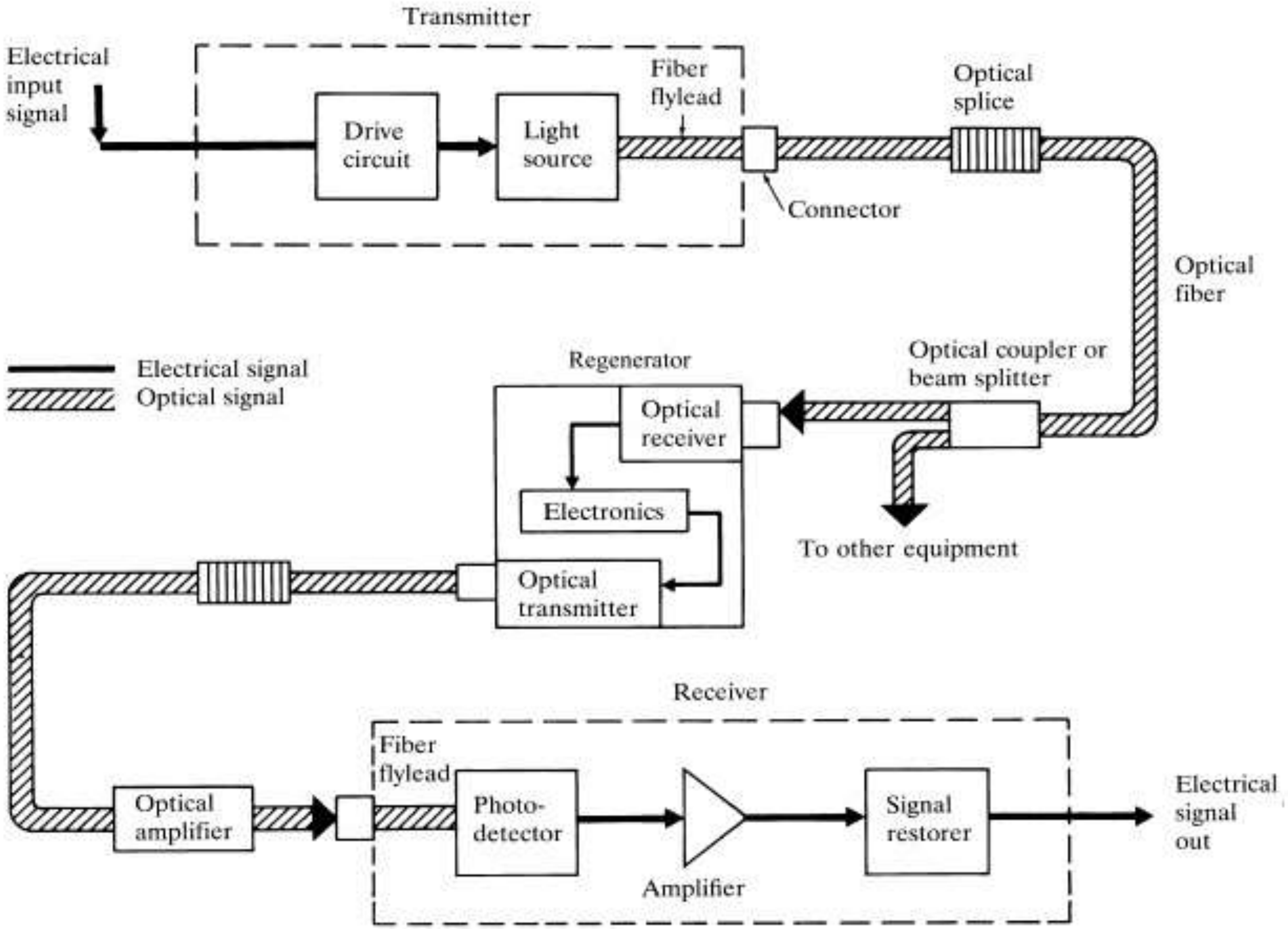


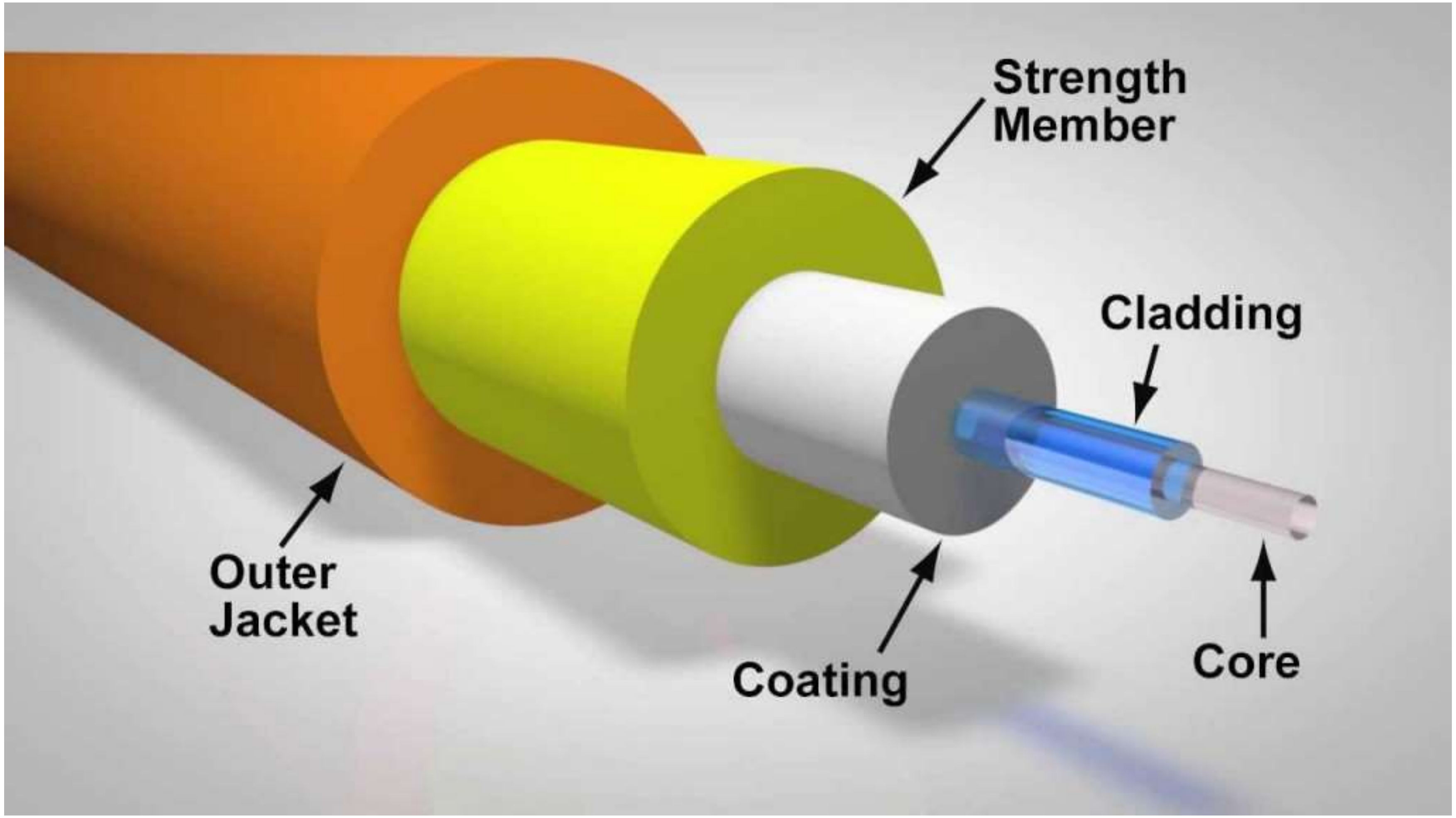
Optical fiber

OF Communication System



OPTICAL FIBER LINK




Strength
Member
Cladding
Outer
Jacket
Coating
Core

Structure of optical fiber

- **<u>CORE</u>**
  - cylindrical rod of dielectric material generally glass.
  - Dielectric material conducts no electricity.
  - Light propagates mainly along the core of the fiber.
  - **core** is described as having a
    - radius of (a)
    - index of refraction n_1.
  - core is surrounded by a layer of material called cladding.

Structure

- **CLADDING**
  - **cladding** layer is made of a dielectric material generally glass or plastic.
  - index of refraction n_2.
- Refractive index of cladding material is less than that of the core material.
  - The cladding performs the following functions:
    - Reduces loss of light from the core into the surrounding air
    - Reduces scattering loss at the surface of the core
    - Adds mechanical strength

Structure…

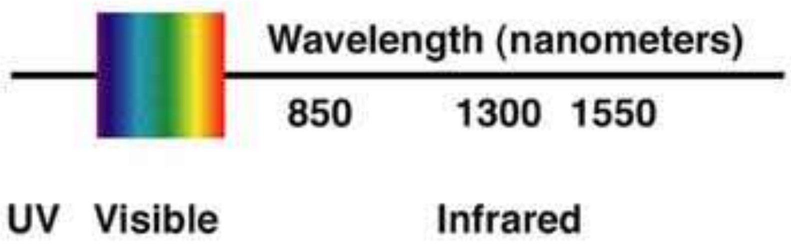
- **COATING/ BUFFER**
  - For extra protection, the cladding is enclosed in an additional layer called the coating or buffer.
  - The coating or buffer is a layer of material used to protect an optical fiber from physical damage.
  - The material used for a buffer is a type of plastic.
  - The buffer is elastic in nature and prevents abrasions.
- **STRENGTHENING MATERIAL**
  - Provides tensile strength
  - Crush resistance
  - Protection from excess bending
  - Vibration isolation

JACKET

protection of fiber inner layers from environmental conditions

OF wavelength

- The three main wavelengths used for fiber optic transmission are 850, 1300, and 1550 nano **meters**.
- Typically multi-mode glass fibers use light at **850 nm** - **1300nm**, referred to as "short wavelength" and single-mode fiber operates at 1310, or **1550 nm**, called "long wavelength".
- Why do we use the infrared? Because the **attenuatio**n(absorption and scattering) of the fiber is much less at those wavelengths. Fortunately, we are also able to make transmitters (lasers or LEDs) and receivers (photodetectors) at these particular wavelengths.
-



*Attenuation mechanisms

The signal on the optical fiber attenuates due to following mechanisms :

1. **Material absorption in the fiber.**

   - **Intrinsic absorption**
     - Infrared
     - UV
   - **Extrinsic absorption**
     - impurities
     - OH molecules
   - **Absorption due to atomic defects**

2. **Scattering due to micro irregularities inside the fiber.**

   Rayleigh and Mie

3. **Bending or radiation losses on the fiber.**

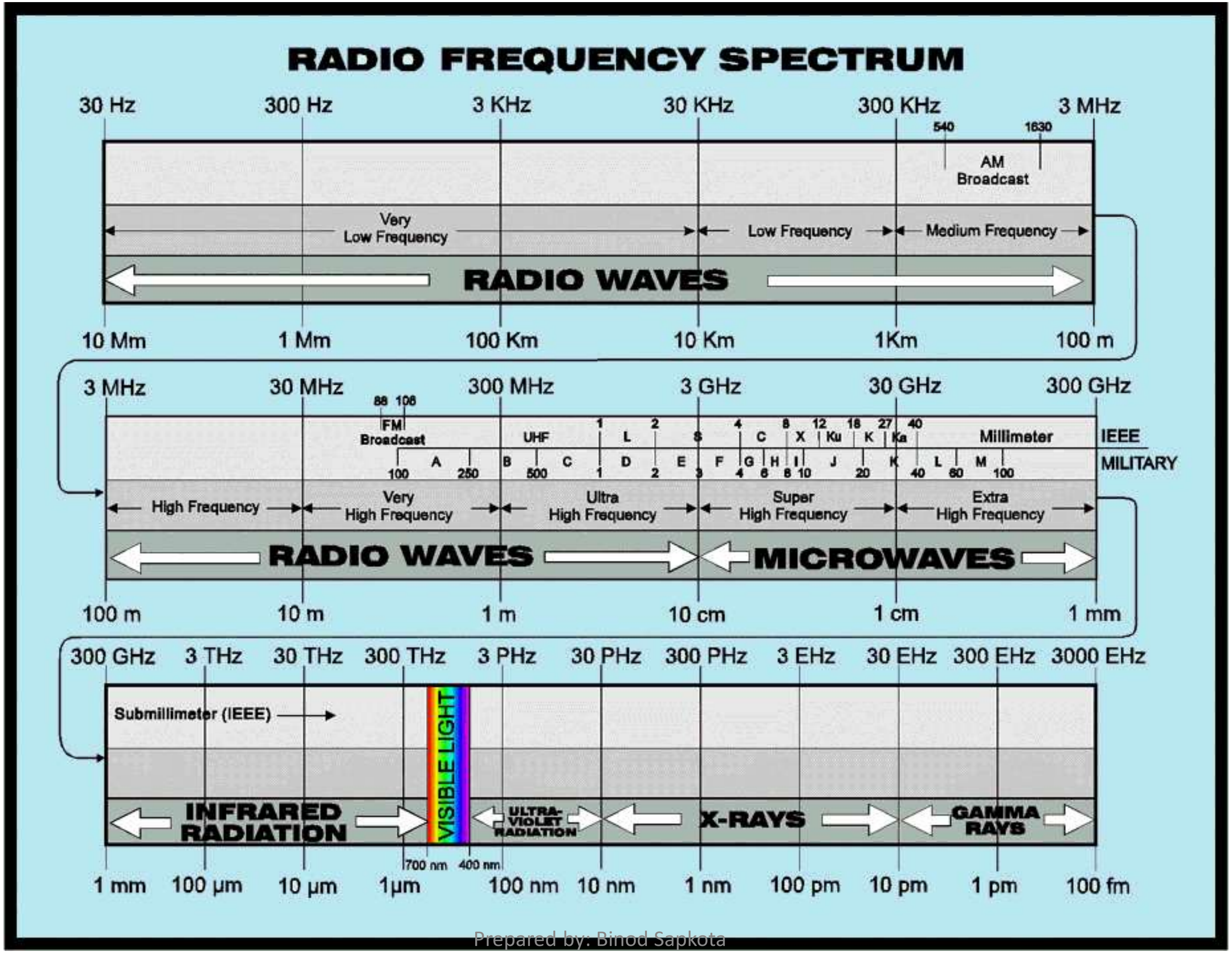
   Macrobending and microbending

The first two losses are intrinsically present in any fiber and the last loss depend on the environment in which the fiber is laid.

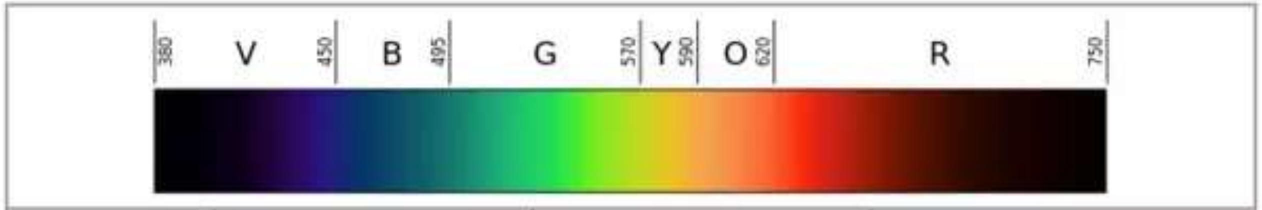
Radio Frequency Spectrum: Ranges

Designation	Abbreviation	Frequencies	Wavelengths
Very Low Frequency	VLF	3 kHz - 30 kHz	100 km - 10 km
Low Frequency	LF	30 kHz - 300 kHz	10 km - 1 km
Medium Frequency	MF	300 kHz - 3 MHz	1 km - 100 m
High Frequency	HF	3 MHz - 30 MHz	100 m - 10 m
Very High Frequency	VHF	30 MHz - 300 MHz	10 m - 1 m
Ultra High Frequency	UHF	300 MHz - 3 GHz	1 m - 100 mm
Super High Frequency	SHF	3 GHz - 30 GHz	100 mm - 10 mm
Extremely High Frequency	EHF	30 GHz - 300 GHz	10 mm - 1 mm

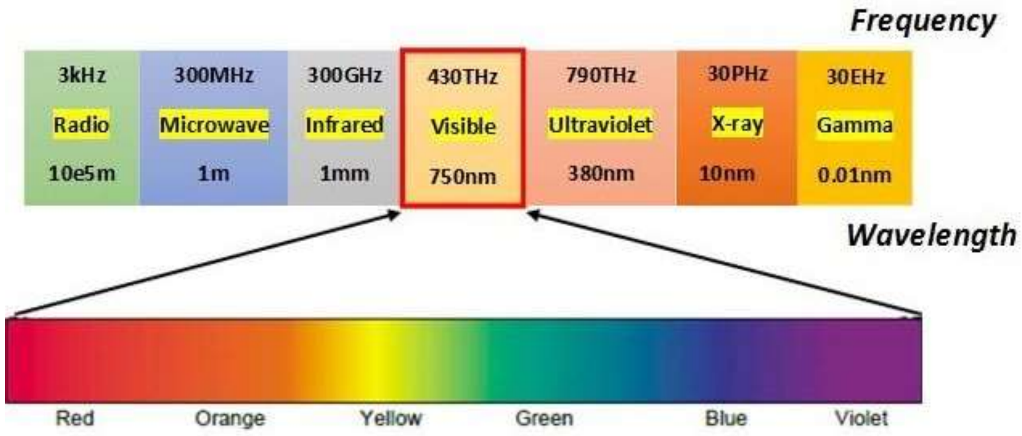
www.rfpage.com



RADIO FREQUENCY SPECTRUM
30 Hz
300 Hz
3 KHz
30 KHz
300 KHz
3 MHz
540
1630
AM Broadcast
Very Low Frequency
Low Frequency
Medium Frequency
RADIO WAVES
10 Mm
1 Mm
100 Km
10 Km
1Km
100 m
3 MHz
30 MHz
300 MHz
3 GHz
30 GHz
300 GHz
88 108
FM Broadcast
UHF
L
S
C
X
Ku
K
Ka
Millimeter
IEEE
MILITARY
A
B
C
D
E
F
G
H
I
J
K
L
M
High Frequency
Very High Frequency
Ultra High Frequency
Super High Frequency
Extra High Frequency
RADIO WAVES
MICROWAVES
100 m
10 m
1 m
10 cm
1 cm
1 mm
300 GHz
3 THz
30 THz
300 THz
3 PHz
30 PHz
300 PHz
3 EHz
30 EHz
300 EHz
3000 EHz
Submillimeter (IEEE)
INFRARED RADIATION
VISIBLE LIGHT
ULTRA-VIOLET RADIATION
X-RAYS
GAMMA RAYS
700 nm
400 nm
1 mm
100 μm
10 μm
1μm
100 nm
10 nm
1 nm
100 pm
10 pm
1 pm
100 fm



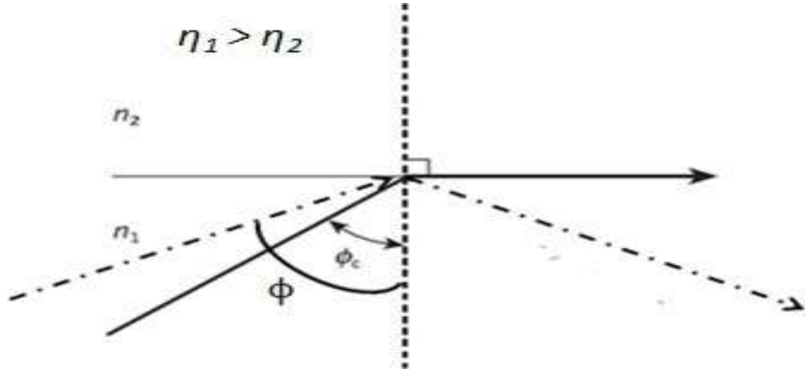
Color	Wavelength	Frequency	Photon energy
violet	380–450 nm	668–789 THz	2.75–3.26 eV
blue	450–495 nm	606–668 THz	2.50–2.75 eV
green	495–570 nm	526–606 THz	2.17–2.50 eV
yellow	570–590 nm	508–526 THz	2.10–2.17 eV
orange	590–620 nm	484–508 THz	2.00–2.10 eV
red	620–750 nm	400–484 THz	1.65–2.00 eV



why this is the range of colors we can see, it's because it is the same region as the brightest output of the sun.

Principle of light propagation in fiber

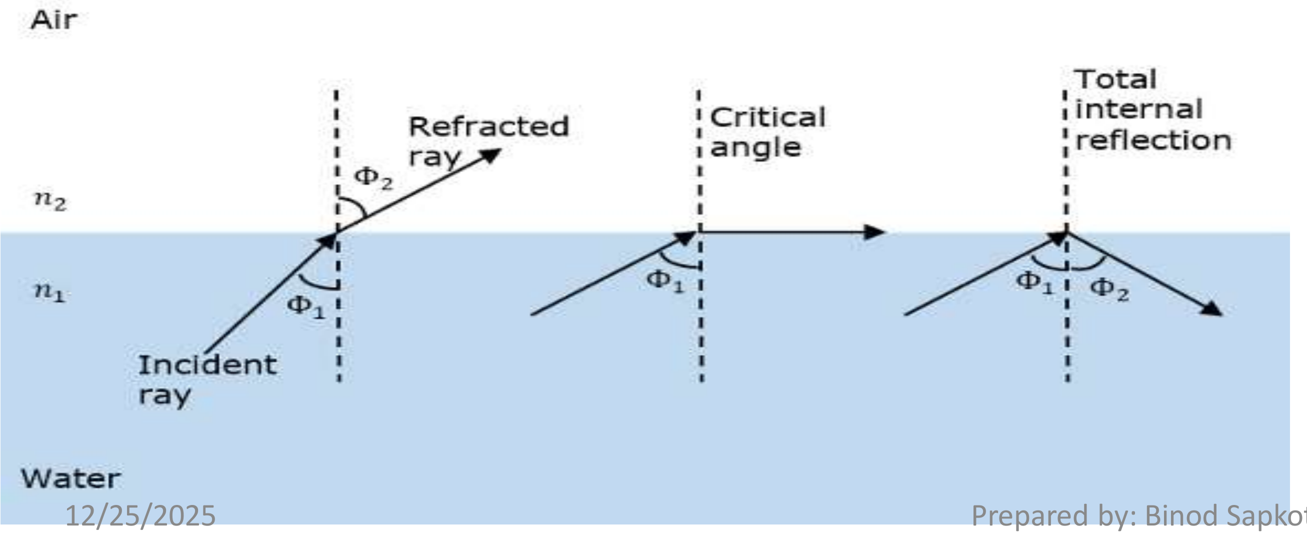
- Total internal reflection (TIR) at core-cladding interface.
- Refractive index of core η_1 is greater than cladding η_2
- When light travels from denser core towards the rarer cladding at an angle ϕ , such that $\phi > \phi_c$, then it is totally reflected back inside the core.

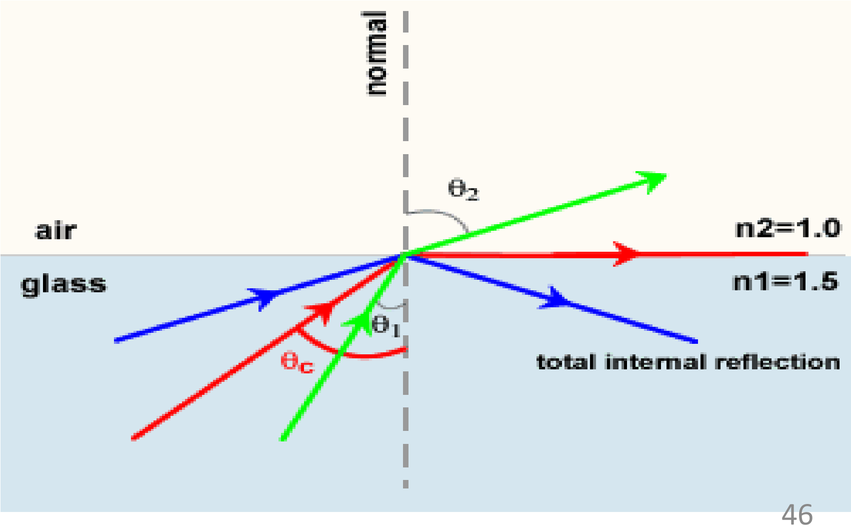


- i.e. when $\phi > \phi_c$, condition of TIR is satisfied.

TOTAL INTERNAL REFLECTION

- By Snell's Law,
- $\eta_1 \text{ Sin } \phi_i = \eta_2 \sin \phi_r$ or, $\eta_1 \text{ Sin } \phi_c = \eta_2 \sin 90^\circ$ *or,*
  $\phi_c = \sin^{-1}(\eta_2 / \eta_1)$
- For TIR, incidence angle $\phi >= \phi_c$





Types of Optical fiber

- On the basis of core-cladding refractive index profile

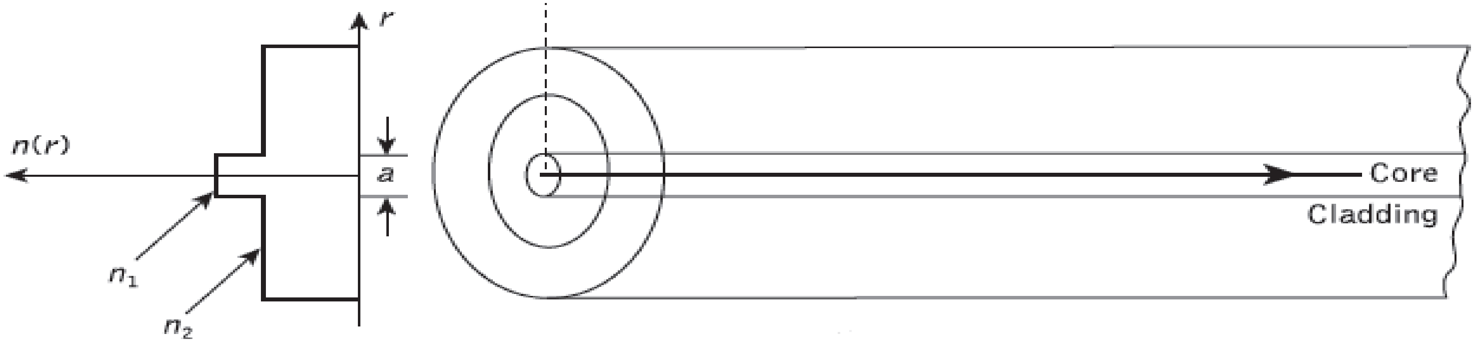
a) Step Index
b) Graded Index

- On the basis of number of mode supported

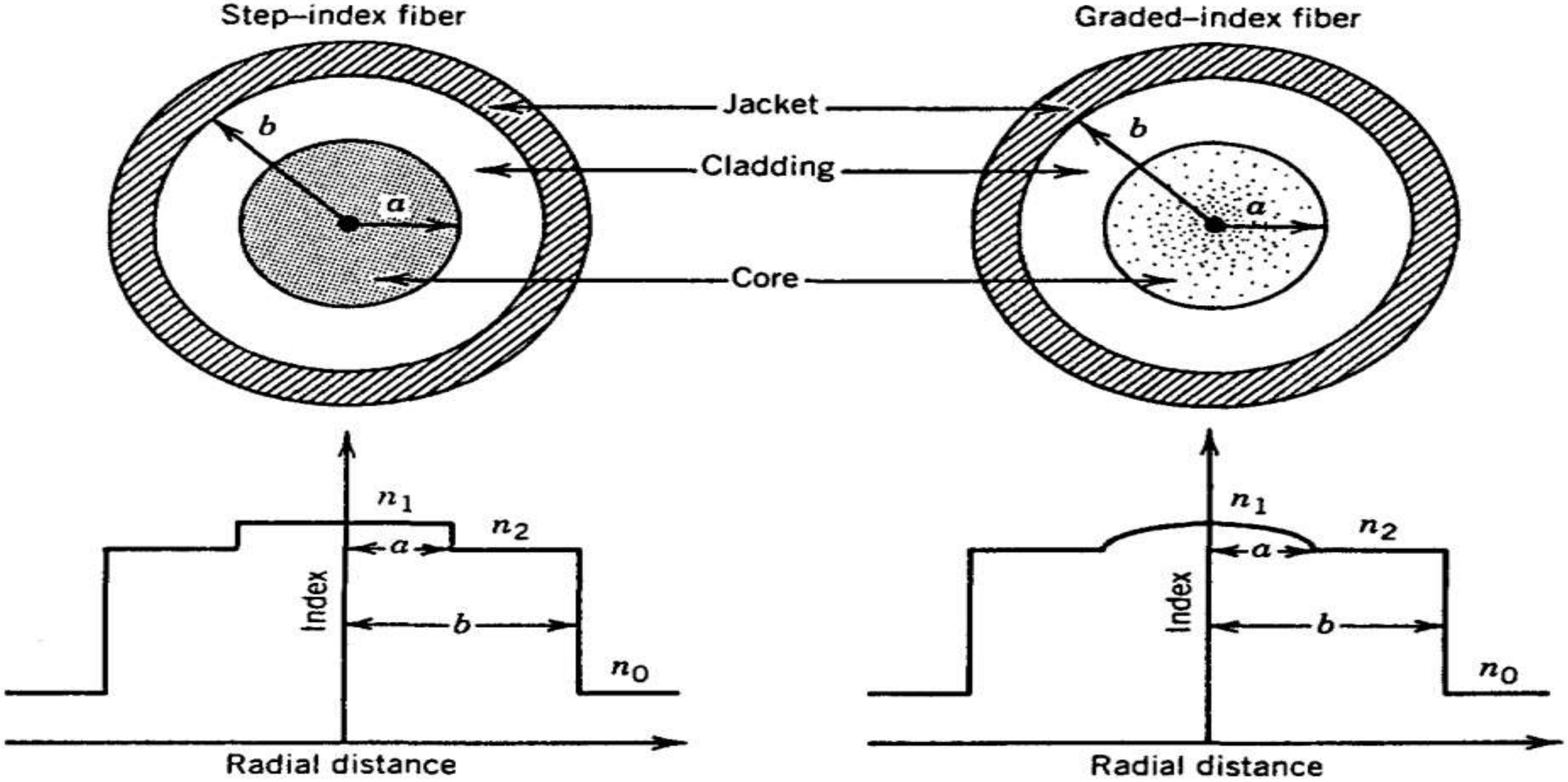
a) Single mode
    - only one signal can be transmitted
    - use of single frequency
b) Multi Mode
    - Several signals can be transmitted
    - Several frequencies used to modulate the signal

Step Index fiber

- Optical fiber with a core of constant refractive index n_1 and a cladding of a slightly lower refractive index n_2 is known as step index fiber.



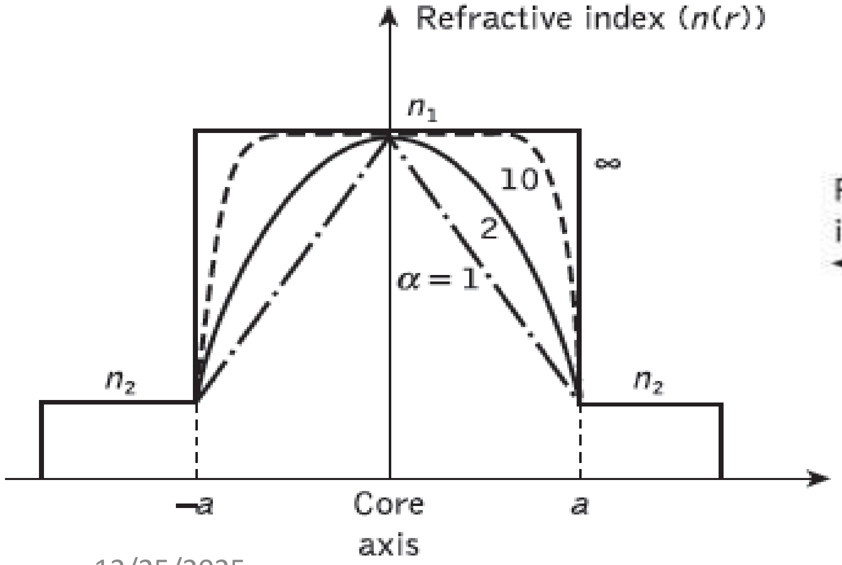
$$n(r) = \begin{cases} n_1 & r < a \quad \text{(core)} \\ n_2 & r \geq a \quad \text{(cladding)} \end{cases}$$

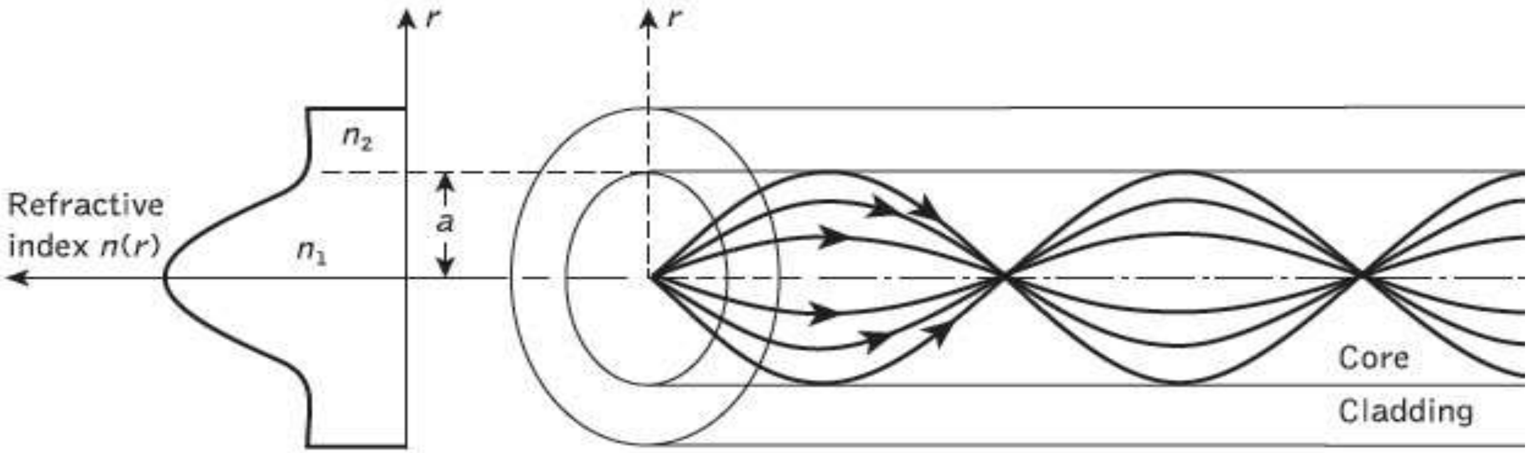


Cross section and refractive-index profile for step-index and graded-index fibers.

Graded Index Fiber

- Graded index fibers have a decreasing core refractive index with radial distance having:
  - maximum value of n_1 at the axis
  - minimum near core cladding interface





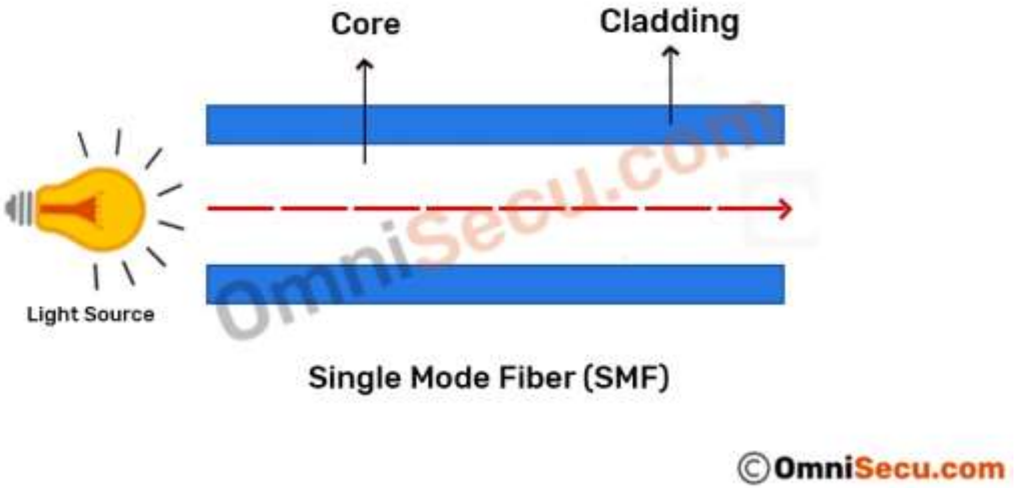
Single Mode Fiber (SMF)

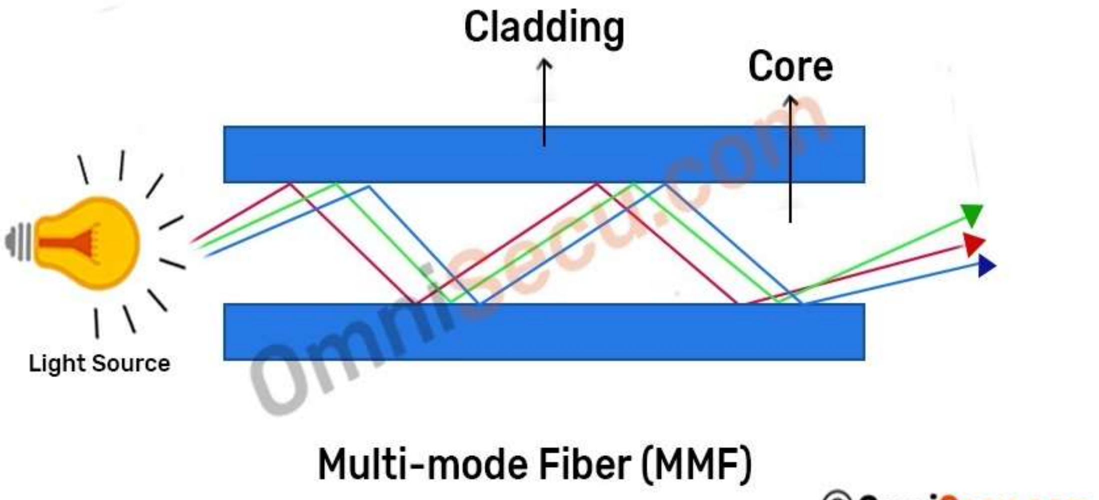
- A single mode or mono mode step index fiber allows the propagation of only one traverse electromagnetic mode and hence the core diameter must be of the order of 2 µm to 10µm.
- Low attenuation and zero Dispersion.
  - Comparatively reduced attenuation is the main advantage of single mode fiber (SMF). Attenuation is the reduction in the intensity of a light signals (thus quality of light signals) when signals are transferred over long distances because of scattering of light.
- Higher data rate
- Suitable for long distance transmission
- It requires highly directional source such as LASER diode because of small core diameter, hence the cost is higher than multimode step index fiber.
- SMFs have a small core size less than 10 microns (µm) compared with multimode optical fibers.

Multimode Fiber (MMF)

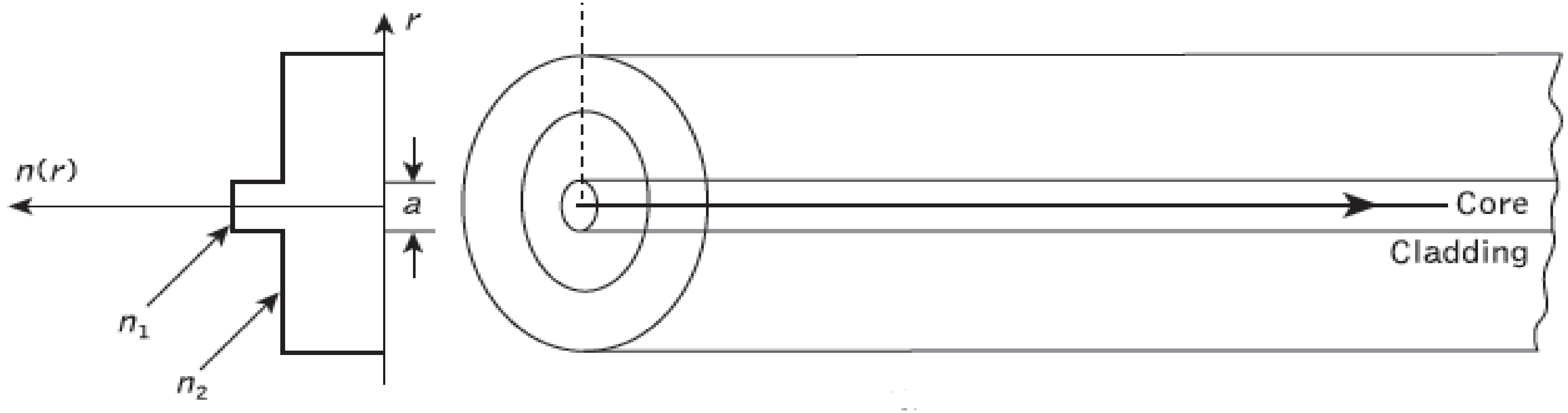
- Has larger core size that guides many mode or rays simultaneously
- MMF cables have large cores (50 µm or 62.5 µm) when compared with single mode optical fibers (less than 10 µm). Multimode fibers can transfer many modes of light signals simultaneously.
- When one pulse of signal is generated into MMF the multiple modes enter the fiber core from different angles and each modes propagates at different speeds and causes pulse broadening(modal dispersion)
- Increased diameter and more modes can cause more light reflections and more attenuation of light signals.
- Generally used for short distance.

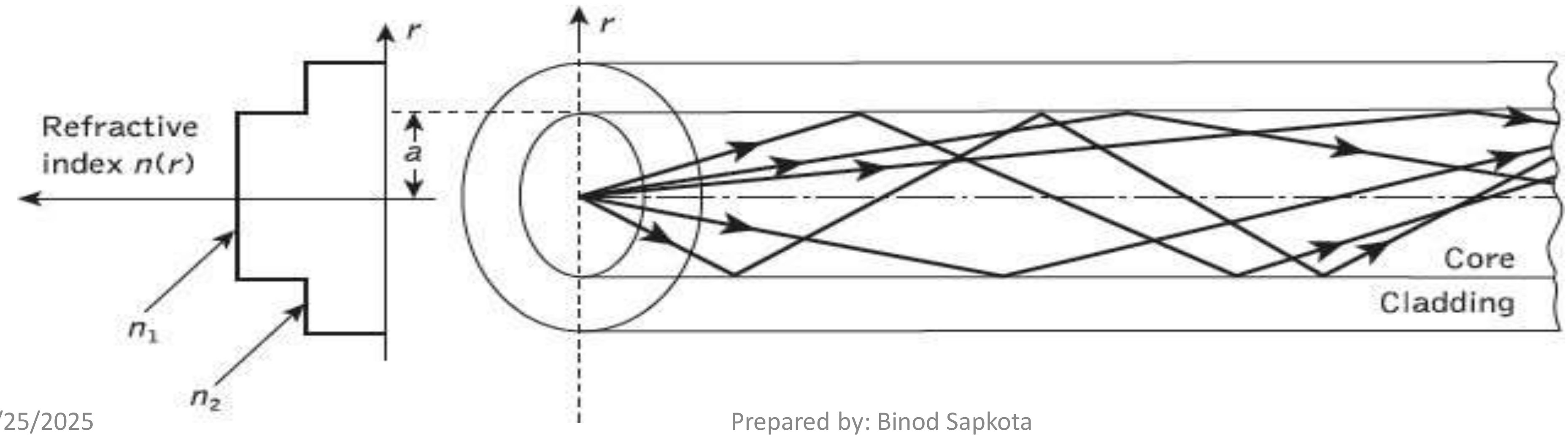
SMF and MMF


Core
Cladding
Light Source
OmniSecu.com
Single Mode Fiber (SMF)
©OmniSecu.com


Cladding
Core
Light Source
OmniSecu.com
Multi-mode Fiber (MMF)
©OmniSecu.com

SMF and MMF


r
n(r)
a
Core
Cladding
n1
n2


r
r
Refractive
index n(r)
a
Core
Cladding
n1
n2

Advantage of MMF

- Light source can be incoherent i.e large number of spectral components.
- Larger Numerical aperture (NA) so easy coupling of light.
- Lower tolerance required in fiber joints.

- MMF cables can transmit data at 100 Mbps (megabits per second) for distances up to 2 kilometers (100Base-FX), 1 Gbps up to 1 kilometer, and 10 Gbps up to 550 meters.

Different multi mode fiber(MMF) types

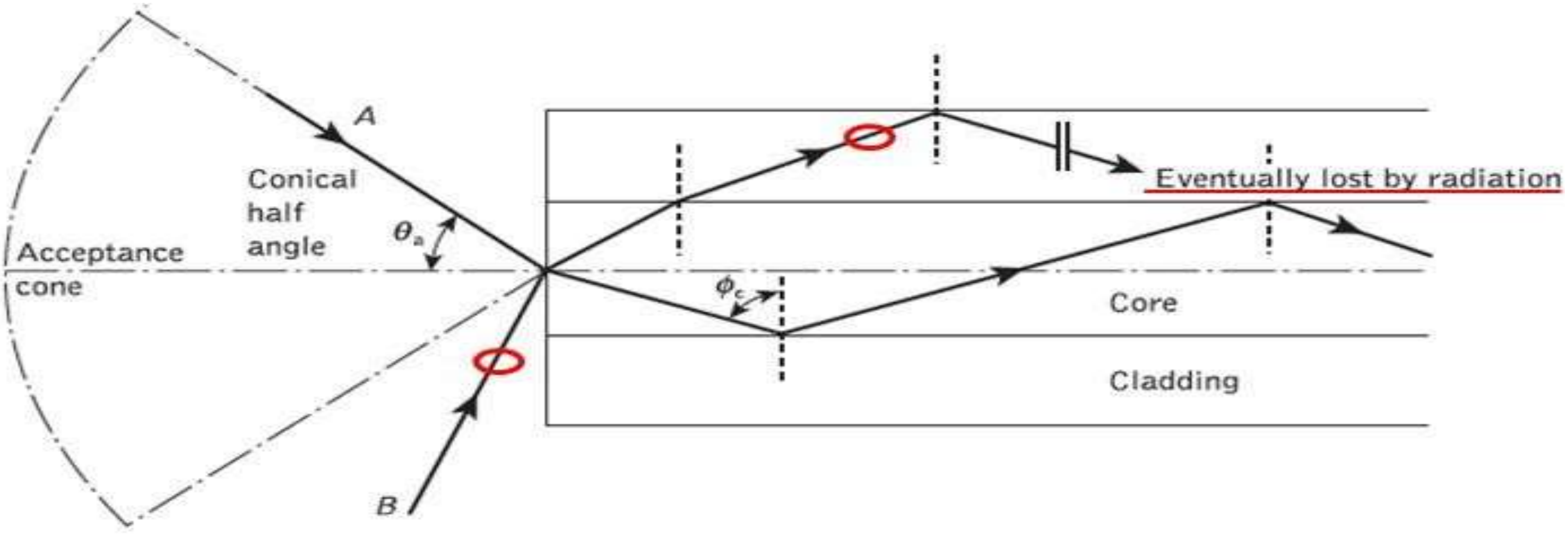
OM Type	Color	Core Size	Speed	Distance	Use
OM1	Orange	62.5µm	1Gbps	300 meters	LANs
OM2	Orange	50.0µm	1Gbps	600 meters	LANs
OM3	Aqua	50.0µm	10Gbps	300 meters	Campus networks
OM4	Aqua	50.0µm	10Gbps	550 meters	Data centers
OM5	Lime green	50.0µm	100Gbps	150 meters	Data centers

Signal Transmission Approach

1. Geometrical Approach (Ray Theory)
2. Electromagnetic Approach (Mode Theory)

- Ray theory is used to approximate the light acceptance and guiding properties of optical fiber.
- Mode theory is used to approximate light as an Electromagnetic wave.

Acceptance angle (Θ_a)

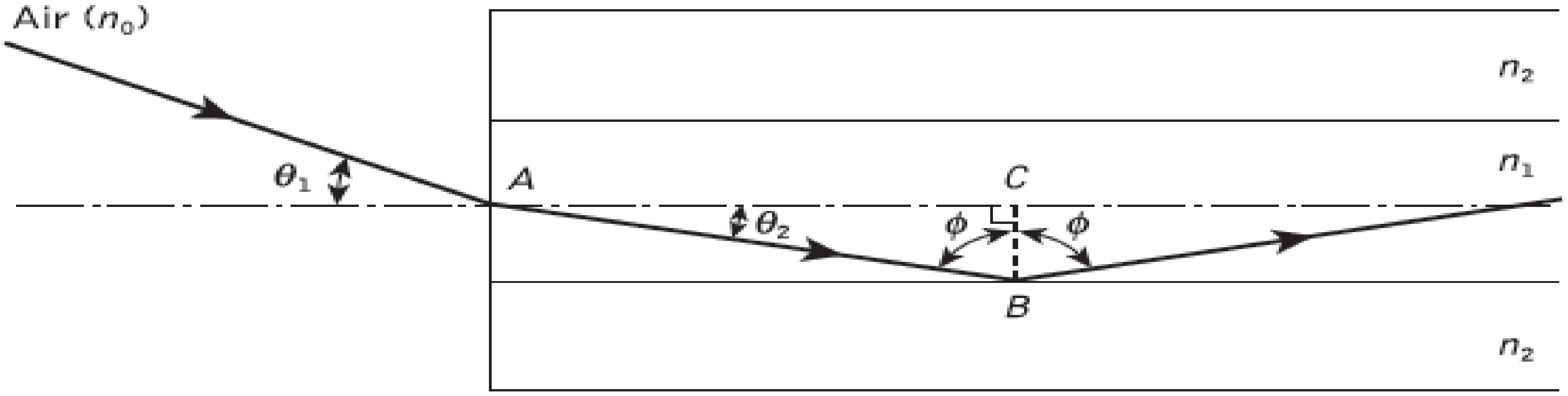


acceptance angle θ_a when launching light into an optical fiber

Acceptance angle of a fiber (Θ_a) is the Maximum value of incidence angle in air that will guide the refracted ray into the core by TIR

Numerical Aperture (NA)

- Light gathering capacity



- Here, consider a meridional ray incident on the fiber core at an angle θ_1 to the fiber axis such that $\theta_1 < \theta_a$
- ray enters the fiber from a medium (air) of refractive index n_0 and $n_1 > n_2$.
- considering the refraction at the air–core interface and using Snell's law,
  - $n_0 \sin \theta_1 = n_1 \sin \theta_2$

NA…

- Or, $n_0 \sin \theta_1 = n_1 \sin (90\text{-}\phi)$ (since $\theta_2 = 90\text{-}\phi$)
- Or, $n_0 \sin \theta_1 = n_1 \cos\phi$
- Or, $n_0 \sin \theta_1 = n_1 (1\text{-} \sin^2\phi)^{1/2}$

For TIR, if $\phi >= \phi_c$, then $\theta_1 = \theta_a$

By Snell's law, $\sin \phi_c = \eta_2 / \eta_1$

So, $n_0 \sin \theta_a = n_1 (1\text{-} (\eta_2 / \eta_1)^2)^{1/2}$

Putting $n_0 = 1$ (for air) and solving

we get, $\sin \theta_a = (n_1^2 - n_2^2)^{1/2} = NA$

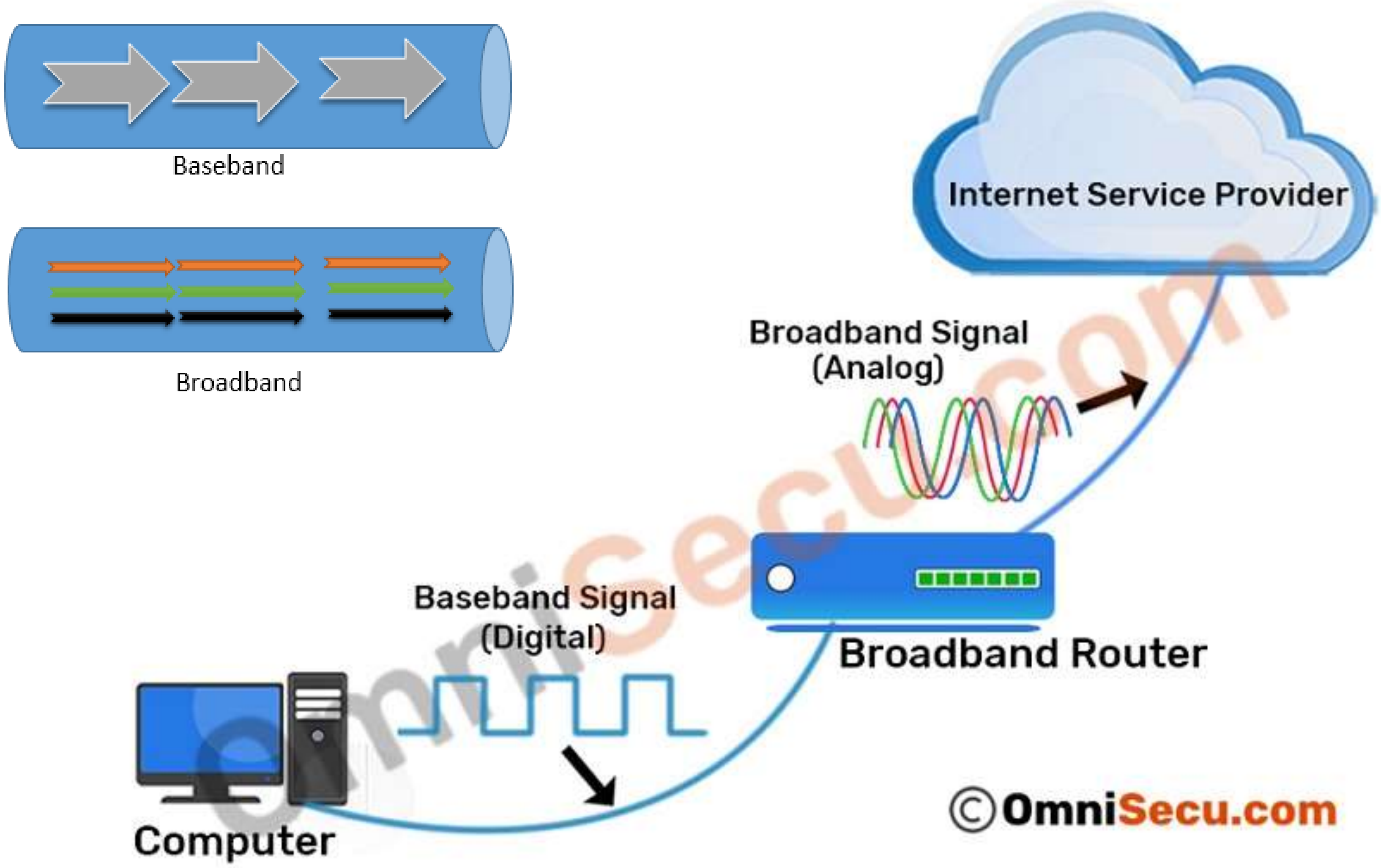
NA is defined as sine of acceptance angle.

Advantages of Optical Fiber

- **Greater Bandwidth & Faster Speed**—Optical fiber cable supports extremely high bandwidth and speed. The amount of information that can be transmitted per unit of optical fiber cable is its most significant advantage.
- **Better Reliability/light signals -**Fiber is immune to temperature changes, severe weather and moisture, all of which can hamper the connectivity of copper cable. Plus, fiber does not carry electric current, so it's not bothered by electromagnetic interference (EMI) that can interrupt data transmission. It also does not present a fire hazard like old or worn copper cables can.
- **Thinner and Light-weighted—**Optical fiber is thinner, and can be drawn to smaller diameters than copper wire. They are of smaller size and light weight than a comparable copper wire cable, offering a better fit for places where space is a concern.
- **Higher carrying capacity—**Because optical fibers are much thinner than copper wires, more fibers can be bundled into a given-diameter cable. This allows more phone lines to go over the same cable or more channels to come through the cable into your cable TV box.
- **Less signal degradation—**The loss of signal in optical fiber is less than that in copper wire.
- **Long Lifespan**—Optical fibers usually have a longer life cycle for over 100 years.
- ***Cheap**—Several miles of optical fiber cable can be made cheaper than equivalent lengths of copper wire.

Disadvantages of Optical Fiber

- **Fragility—**Optical fiber is rather fragile and more vulnerable to damage compared to copper wires. You'd better not to twist or bend fiber optic cables.
- **Difficult to Install-** it's not easy to splice fiber optic cable. And if you bend them too much, they will break. And fiber cable is highly susceptible to becoming cut or damaged during installation or construction activities. All these make it difficult to install.
- **Attenuation & Dispersion**: as transmission distance getting longer, light will be attenuated and dispersed, which requires extra optical components like EDFA to be added.
- ***Limited Application—**Fiber optic cable can only be used on ground, and it cannot leave the ground or work with the mobile communication.
- ***Low Power—**Light emitting sources are limited to low power. Although high power emitters are available to improve power supply, it would add extra cost.
- ***Distance—**The distance between the transmitter and receiver should keep short or repeaters are needed to boost the signal.


Baseband
Broadband
Internet Service Provider
Broadband Signal
(Analog)
Baseband Signal
(Digital)
Broadband Router
Computer
©OmniSecu.com

Baseband

- In Baseband, data is sent as digital signals through the media as a single channel that uses the entire bandwidth of the media.
- Baseband communication is bi-directional, which means that the same channel can be used to send and receive signals.
- Baseband supports the Time Division Multiplexing (TDM).
- Baseband is used for short distance data transfer. Entire bandwidth of the cable is used for single signal transmission. Common Ethernet Standards use Baseband for LAN data transfer.
- If you are using a broadband internet connection for your home internet, the signals from your internet service provider up to your broadband router are broadband signals. But, the signals used inside your Ethernet Local Area Network (LAN) are baseband signals.

Baseband Transmission

- Signal Type: Digital.
- Channel: Uses the entire bandwidth for a single data stream.
- Multiplexing: Time Division Multiplexing (TDM).
- Direction: Bidirectional (send/receive on same channel).
- Distance: Short distances.
- Examples: Ethernet, LANs.

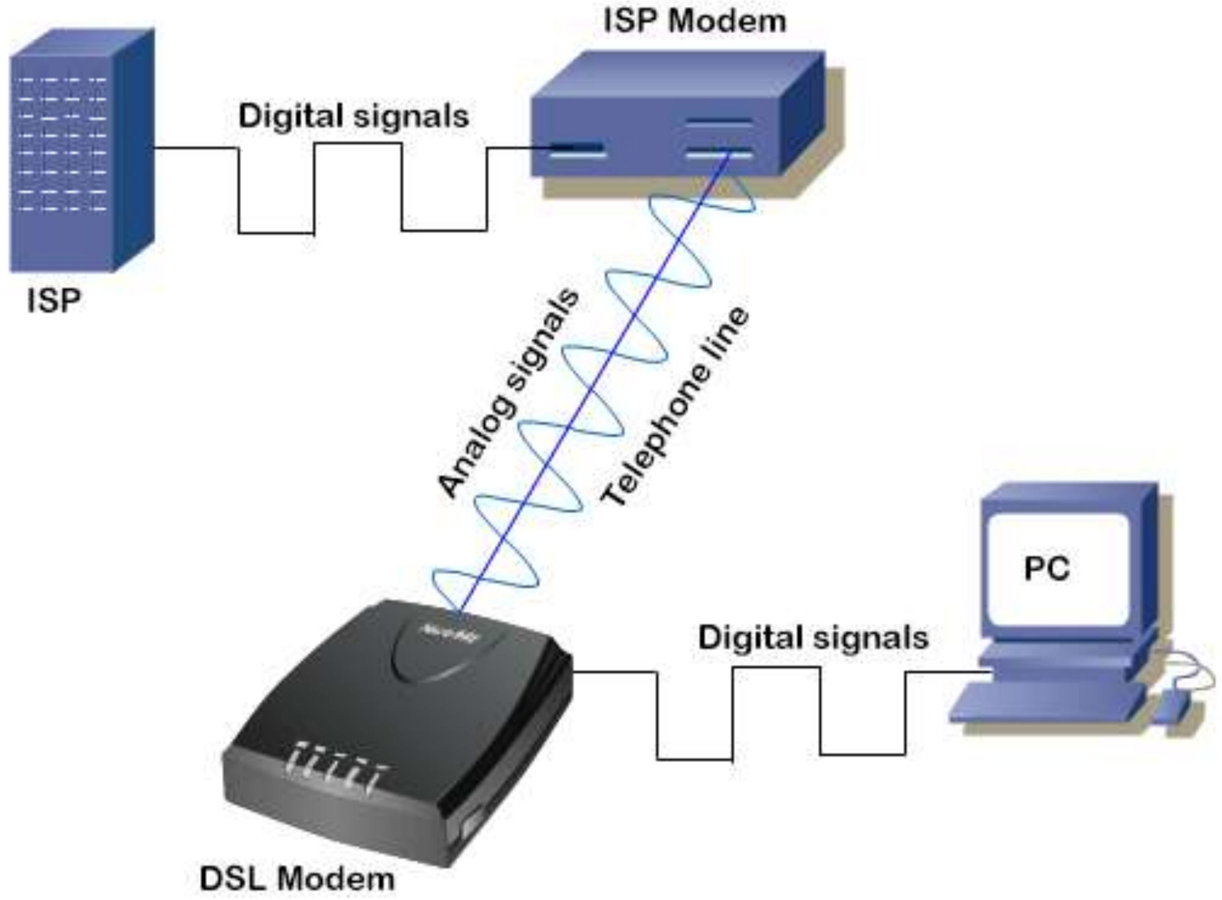
Broadband transmission

- Broadband sends information in the form of an analog signal.
- Each transmission is assigned to a portion of the bandwidth, hence multiple transmissions are possible at the same time.
- Broadband communication is unidirectional, so in order to send and receive, two pathways are needed.
- This can be accomplished either by assigning a frequency for sending and assigning a frequency for receiving along the same cable or by using two cables, one for sending and one for receiving.
- In broadband frequency-division multiplexing is possible.
- Broadband is used for long distance data transfer. Entire bandwidth of the cable is used for multiple signal transmission at different frequencies. Home Internet connection and TV cables use broadband for data transfer.
- **Broadband** refers to various high-capacity **transmission** technologies that are used to **transmit** data, voice, and video across long distances and at high speeds. Common mediums of **transmission** include coaxial cable, fiber optic cable, and radio waves

Broadband transmission

- Broadband Transmission
- Signal Type: Analog.
- Channel: Divided into multiple frequency channels for simultaneous signals.
- Multiplexing: Frequency Division Multiplexing (FDM).
- Direction: Unidirectional (requires separate channels for send/receive).
- Distance: Long distances.
- Examples: Cable TV, DSL, cellular, internet.

Data modulation



Multiplexing, Circuit switching, Packet switching, VC Switching, Telecommunication switching system (Networking of Telephone exchanges)

- (Multiplexing is a process where multiple analog message signals or digital data streams are combined into one signal over a shared medium.)
- Multiplexing in data communications is a method that combines multiple signals or data streams into one signal over a shared medium. This process allows for efficient use of resources and can significantly increase the amount of data that can be sent over a network.

Need of Multiplexing

- To reduce the number of electrical connections or lead.
- To share the bandwidth between the users.
- To increase the capacity of channel.
- To increase the transmission speed.
- To make the signal secure.
- To make scalable.
- To make cost efficiency.

Multiplexing is crucial in communication systems to efficiently share a single, high-capacity channel for multiple low-bandwidth signals, maximizing bandwidth use, reducing costs, and optimizing infrastructure like fiber optics or radio frequencies.
It prevents signal mixing, allows simultaneous data transmission, improves efficiency, and enables scalable networks by combining many streams into one for transmission and separating them at the receiver, saving resources and lowering per-channel costs.

FDM

- This multiplexing technique is used in analog communication.
- This technique works in two steps.
  - In the first step, it divides the communication channel into sub-channels and assigns a separate sub-channel to each node.
  - In the second step, it modulates the frequency of the carrier wave of each node. A carrier wave is a simple analog wave that does not contain any data. A node uses a carrier wave to transmit digital signals over an analog channel.

FDM

FDM is an analog multiplexing technique that combines analog signals.

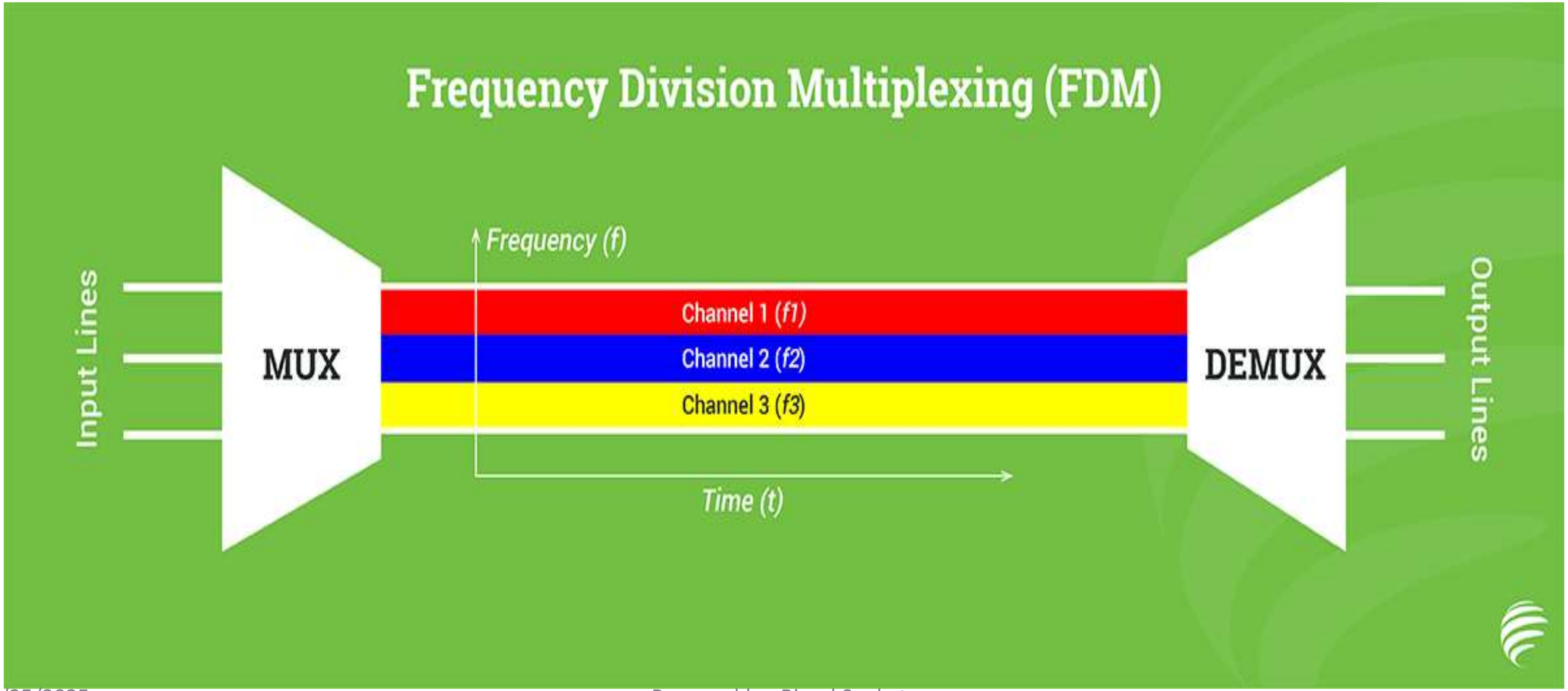


Figure: *FDM process*

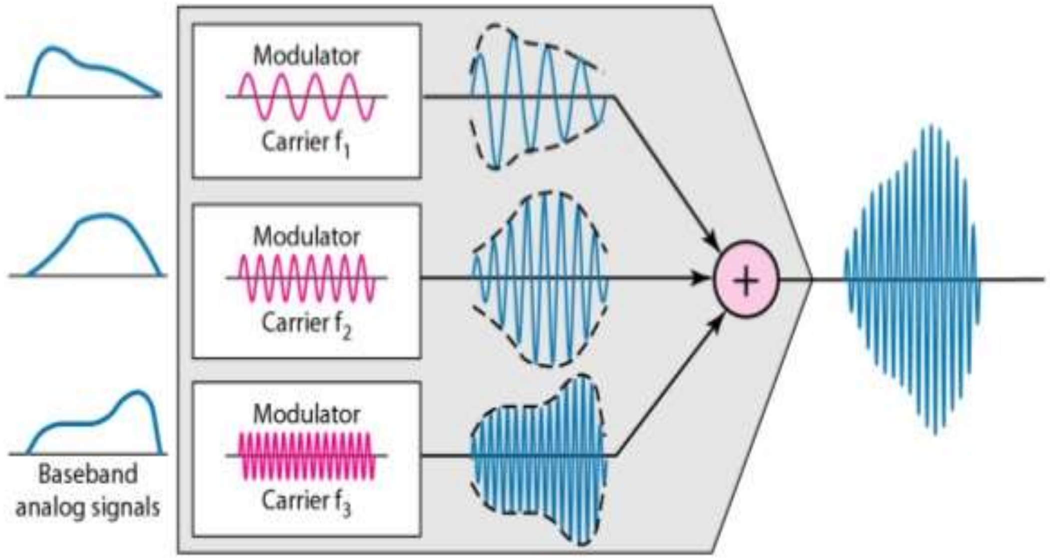


Figure : *FDM demultiplexing example*

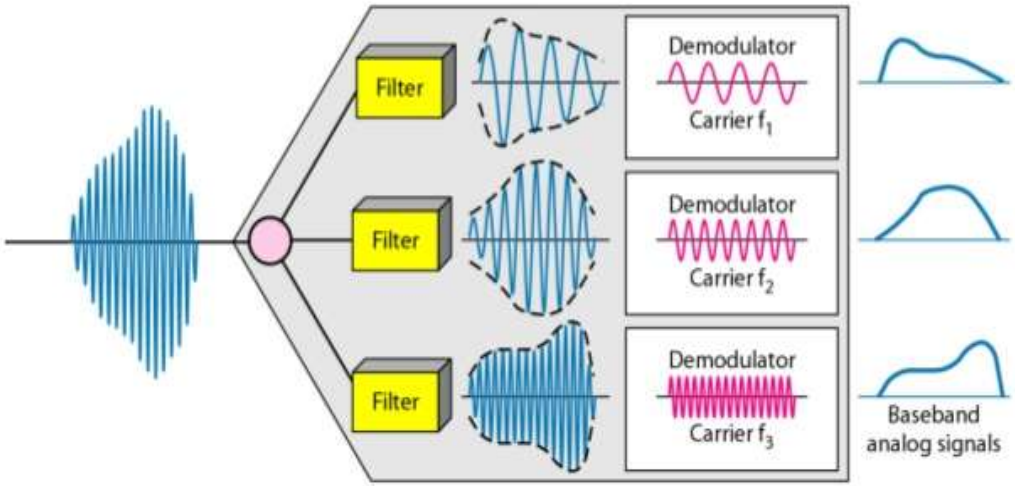
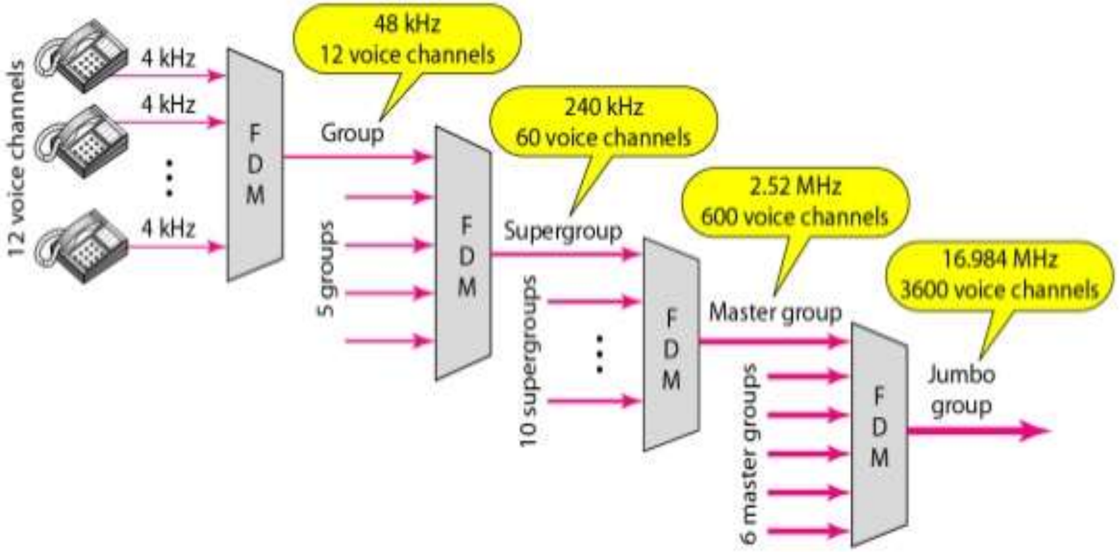


Figure : *FDM hierarchy*



FDM Hierarchy (Analog Hierarchy)

S.N	Designation	Combination	Number of Voice Channel	Bandwidth
1	Voice Channel	1 Voice channel	1	4KHz
2	Group	12 Voice channel	12	48KHz
3	Super Group	5 Group	60	240KHz
4	Master Group	10 Super Group	600	2.52MHz
5	Super Master Group	6 Master Group	3600	17MHz

Advantages and Disadvantages of FDM

Advantages:

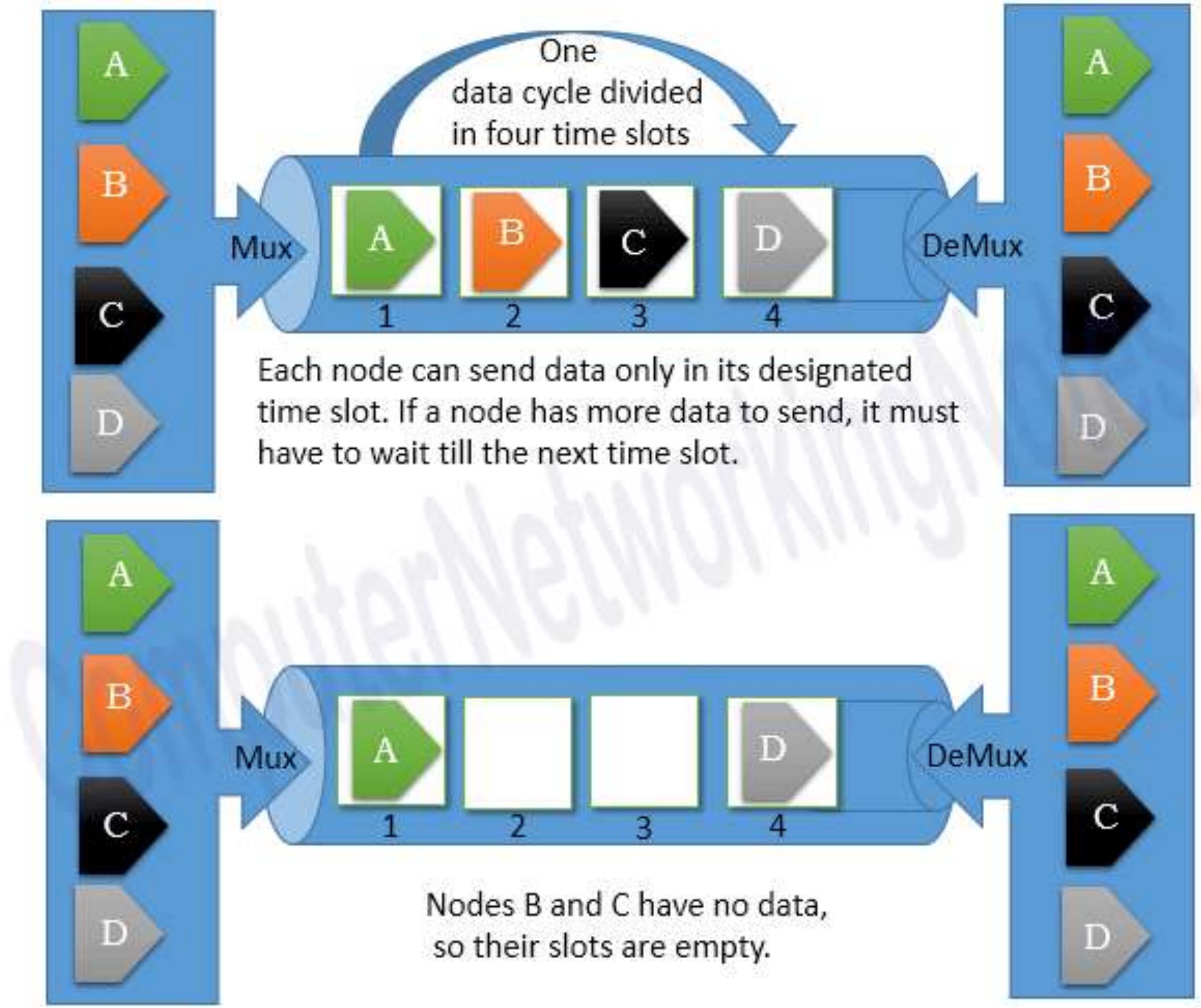
- **Doesn't need synchronization between transmitter and receiver.**
- **Large number of signals (channels) can be transmitted simultaneously.**
- **Demodulation process of FDM is easy.**

Disadvantages:

- **Communication channel must have large bandwidth.**
- **FDM suffers from the problem of crosstalk.**
- **Large numbers of filters and modulators are required.**
- **Intermodulation distortion takes place.**

TDM

- In this technique, time intervals are used to merge signals.
- The multiplexer creates time-slots equal to the number of sending nodes and then assigns a separate time-slot to each node.
- Each node can send data only in its designated time-slot. If a node has no data to send, it sends nothing in its time-slot. If a node has more data to send, it must have to wait till the next time-slot.
- This technique is not more efficient because it reserves a time-slot for each participant node, regardless of whether a participant node has any data to send or not. A node that rarely sends data can waste too much bandwidth by keeping its specified time-slot empty in each data cycle.


One
data cycle divided
in four time slots
A
B
C
D
Mux
DeMux
1
2
3
4
Each node can send data only in its designated time slot. If a node has more data to send, it must have to wait till the next time slot.
Nodes B and C have no data,
so their slots are empty.
ComputerNetworkingNotes

TDM

TDM is a digital multiplexing technique for combining several low-rate channels into one high-rate one.

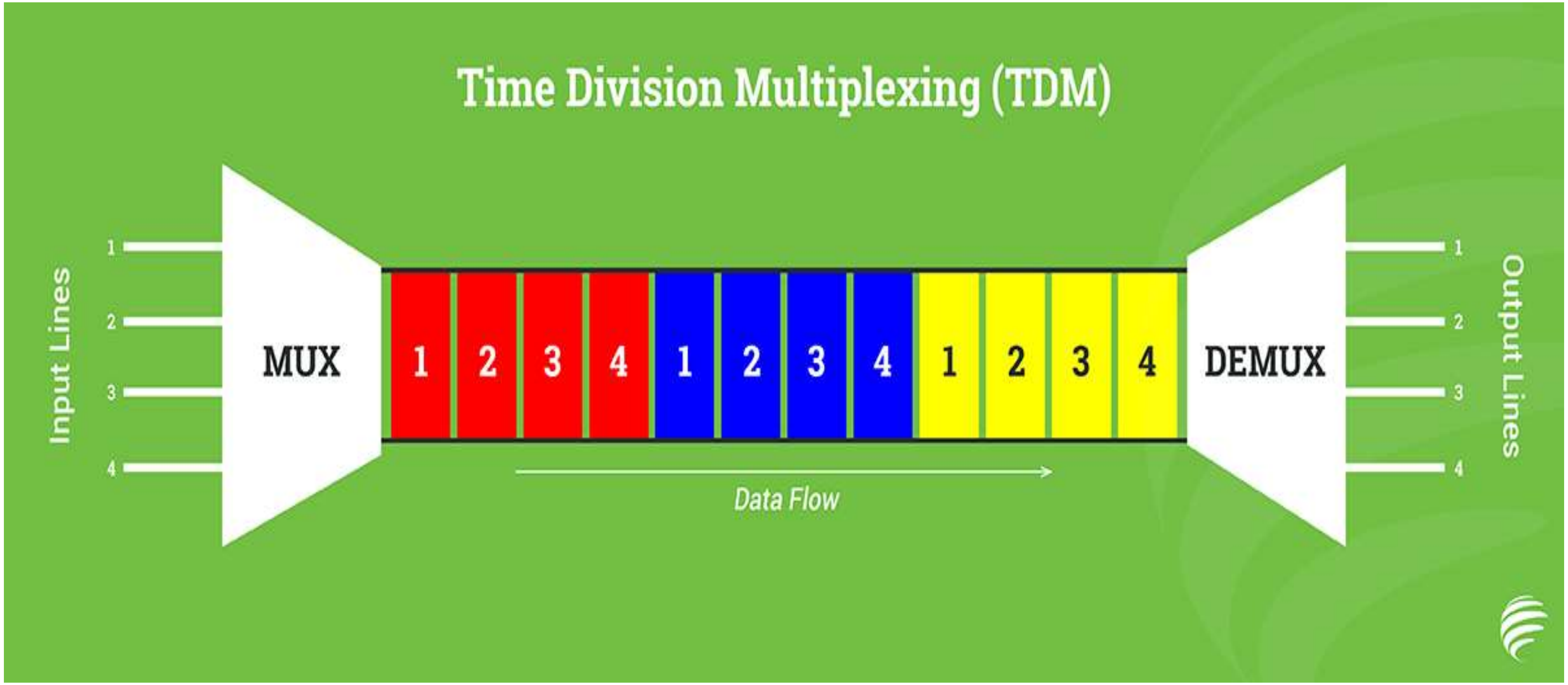
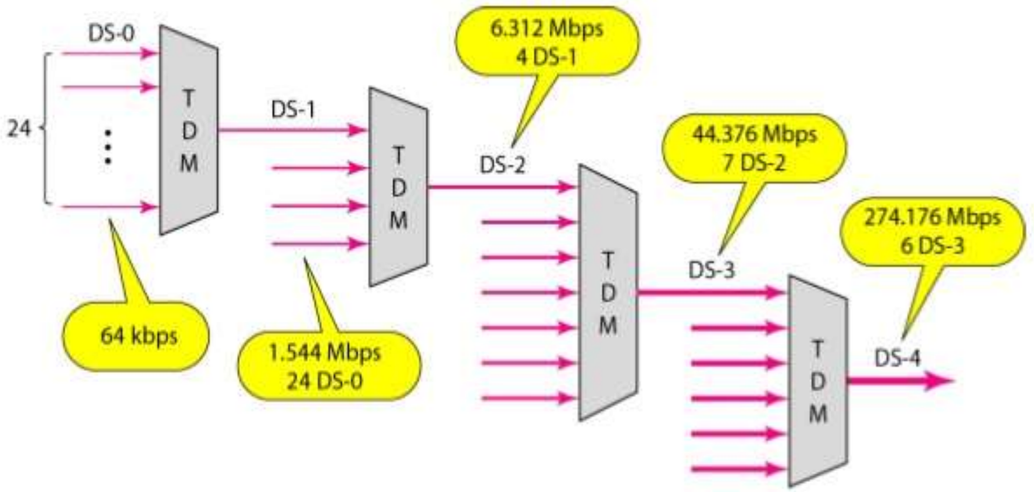


Figure : ***Digital hierarchy- TDM Hierarchy***

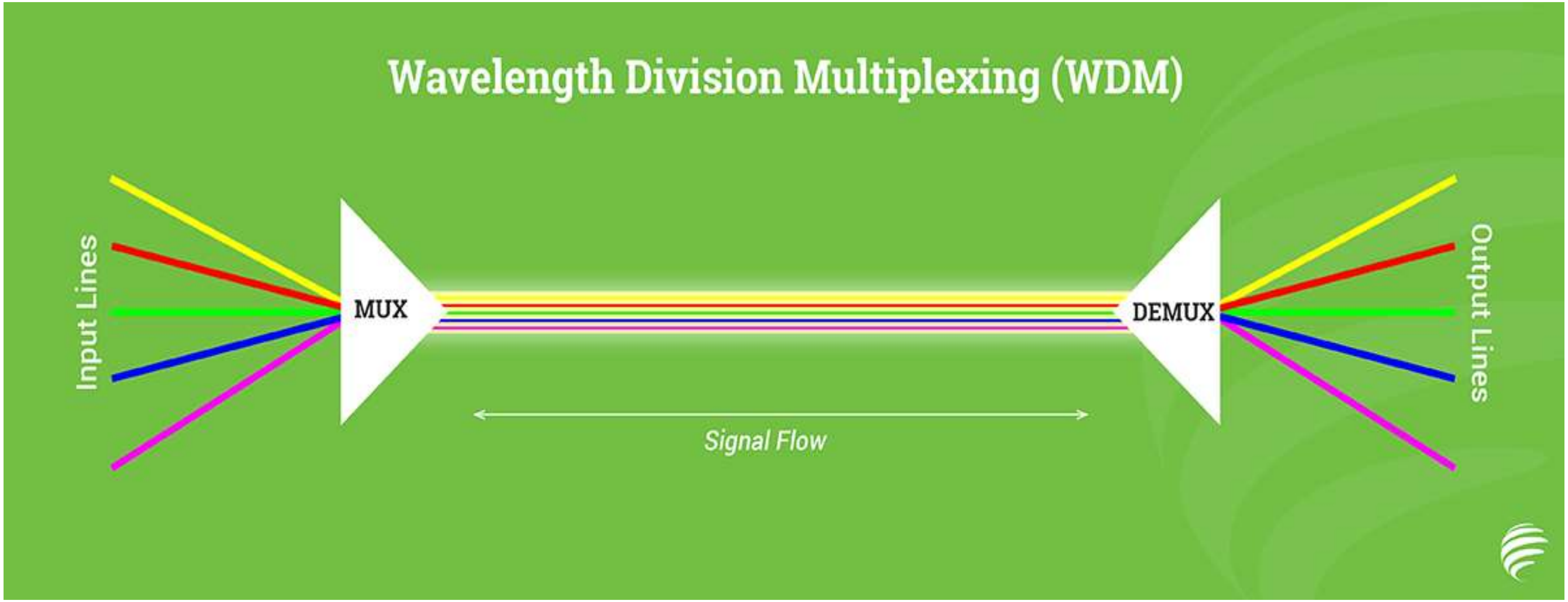


Wavelength Division Multiplexing (WDM)

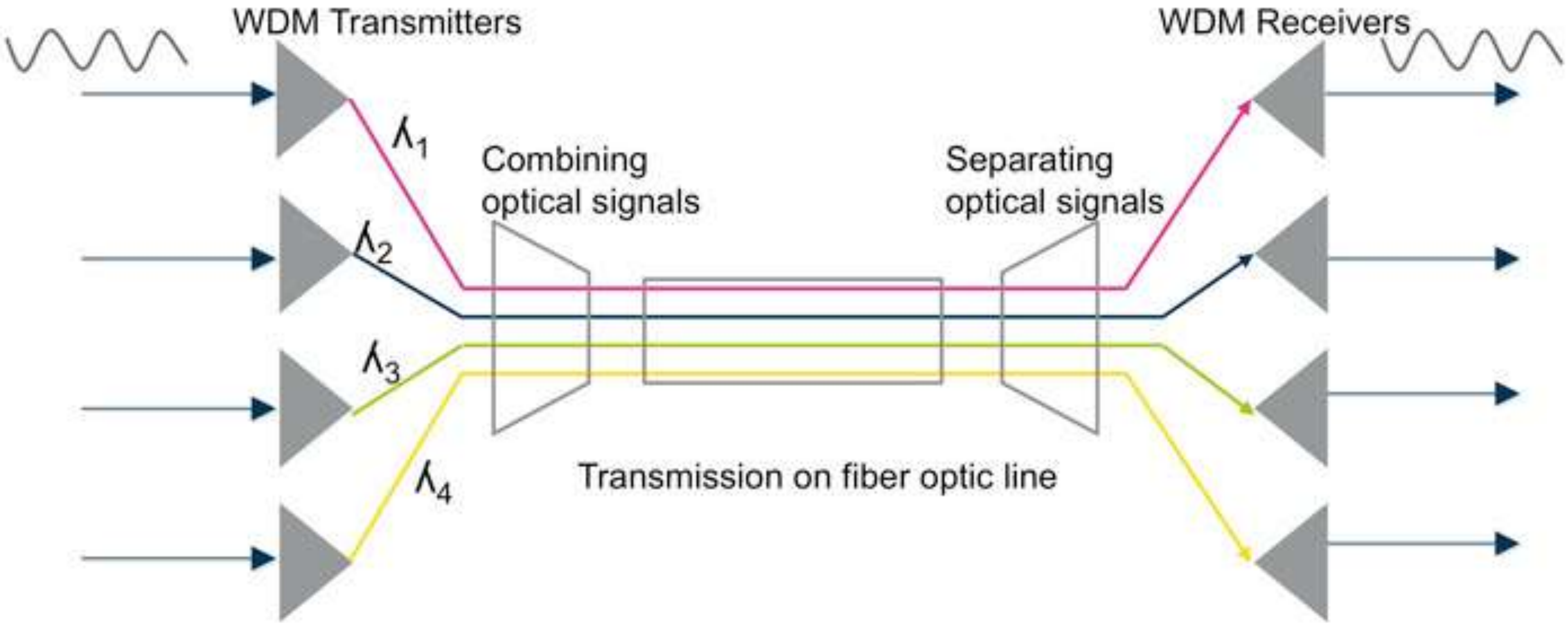
- WDM is a multiplexing technology used to increase the capacity of optical fiber by transmitting multiple optical signals simultaneously over a single optical fiber, each with a different wavelength.
- Each signal is carried on a different wavelength of light, and the resulting signals are combined onto a single optical fiber for transmission.
- At the receiving end, the signals are separated by their wavelengths, demultiplexed and routed to their respective destinations.
- It is used in telecommunications, cable TV, ISPs, and data centers for high-speed, long-distance data transmission.

WDM

WDM is an analog multiplexing technique to combine optical signals.



WDM


WDM Transmitters
WDM Receivers
λ_1
λ_2
λ_3
λ_4
Combining
optical signals
Separating
optical signals
Transmission on fiber optic line

WDM-Types

- WDM can be divided into two categories:
- Dense Wavelength Division Multiplexing (DWDM) is used to multiplex a large number of optical signals onto a single fiber, typically up to 80 (Up to ?) channels with a spacing of 0.8 nm or less between the channels.
- Coarse Wavelength Division Multiplexing (CWDM) is used for lower-capacity applications, typically up to 18 (up to ?) channels with a spacing of 20 nm between the channels.

DWDM

- DWDM uses more wavelengths for signaling than the original form of WDM.
- Because of this, a signal fiber cable can carry between 80 to 160 channels.
- This technology uses costly transceivers equipment.
- Due to cost, usually, this technology is only used on high-bandwidth or long-distance WAN links.

CWDM

- CWDM was developed in an effort to lower the cost of the transceiver equipment.
- This technology uses cheaper transceivers equipment.
- In this technology, channels are spaced more widely and signals are not amplified.
- Because of these, the effective distance of CWDM is less than the original from WDM.
- Through CWDM, a signal fiber cable can carry between 8 to 16 channels.

Why WDM?

- Capacity upgrade of existing fiber networks. (without adding fibers)
- Transparency: Each optical channel can carry any transmission format. (different asynchronous bit rates, analog or digital)
- Scalability– Buy and install equipment for additional demand as needed.
- Wavelength routing and switching: Wavelength is used as another dimension to time and space.

Advantages over Time Division Multiplexing (TDM):

- Higher data rates & capacity
- Lower power consumption
- Reduced equipment complexity
- Flexible & easily upgradable

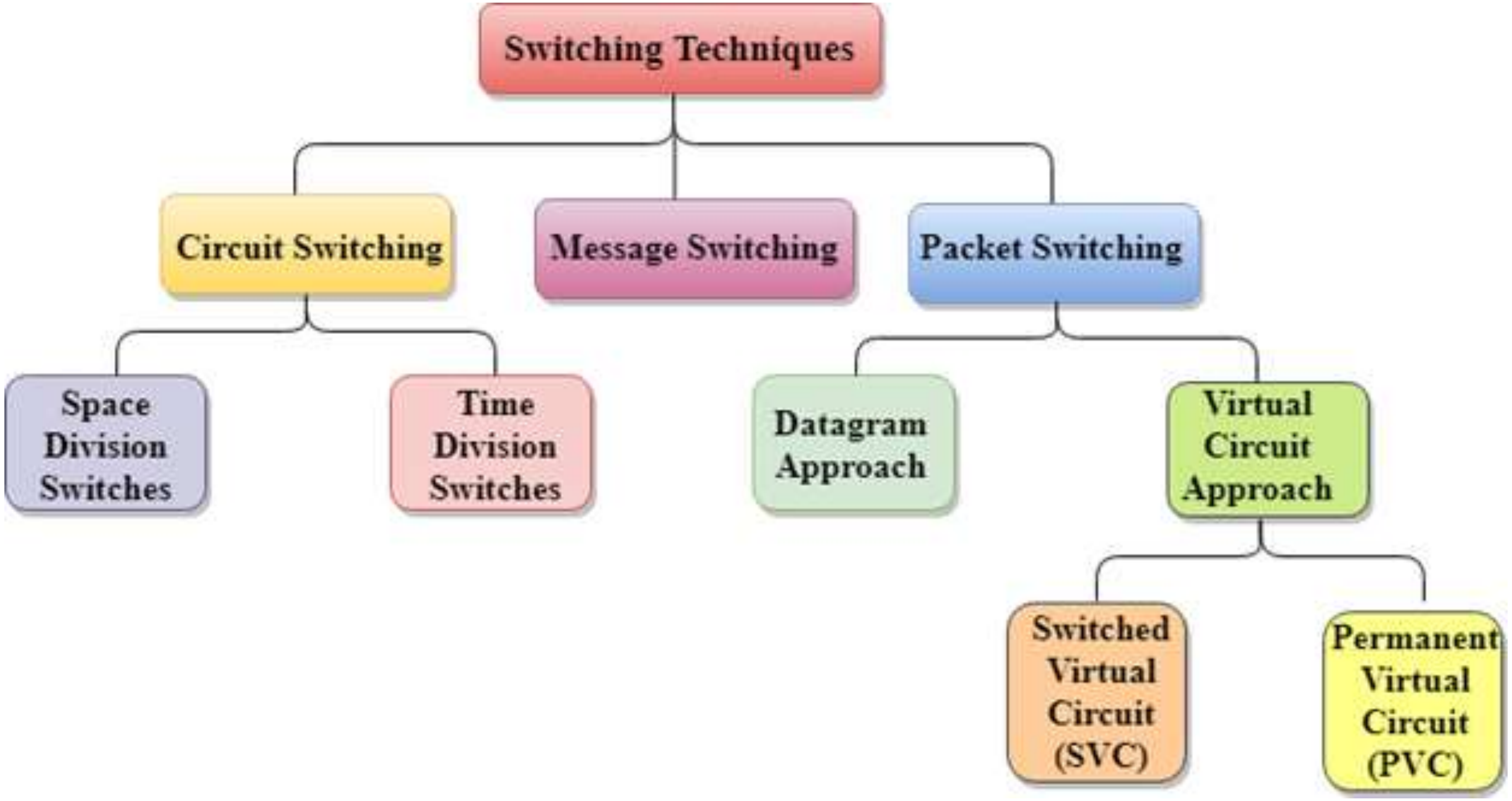
Advantages of Multiplexing

- Efficient Use of Bandwidth: You can send more than one signal over a single channel. This way, you can use the channel's capacity more efficiently.
- Increased Data Transmission: Multiplexing can significantly boost the amount of data that can be sent over a network simultaneously, enhancing overall transmission capacity.
- Scalability: Multiplexing allows networks to easily expand and accommodate more data streams without requiring significant changes to the existing infrastructure.
- Flexibility: Different types of multiplexing (TDM, FDM, WDM, CDM) can be used based on the specific needs and characteristics of the communication system, providing flexibility in network design.

Disadvantages of Multiplexing

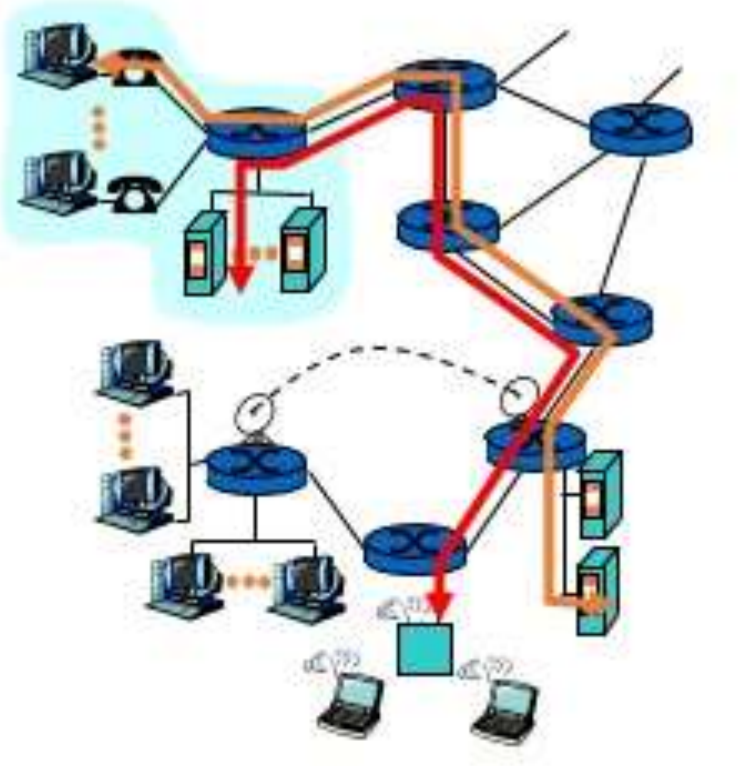
- Synchronization Issues: Ensuring that multiple data streams remain properly synchronized can be challenging, leading to potential data loss or errors if not managed correctly.
- Latency: Combining multiple signals into one can introduce delays, as each data stream needs to be processed, synchronized, and demultiplexed at the receiving end.
- Signal Degradation: Over long distances, multiplexed signals can experience degradation and interference, requiring additional measures such as signal boosters or repeaters to maintain quality.
- Resource Management: Allocating and managing resources for multiplexing can be complicated, requiring careful planning and real-time adjustments to avoid congestion and ensure efficient operation.

Switching: Circuit switching, Packet switching



Circuit Switching

- Circuit switching is a technique that directly connects the sender and the receiver in an unbroken path.
- Telephone switching equipment, for example, establishes a path that connects the caller's telephone to the receiver's telephone by making a physical connection.
- With this type of switching technique, once a connection is established, a dedicated path exists between both ends until the connection is terminated.
- Routing decisions must be made when the circuit is first established, but there are no decisions made after that time

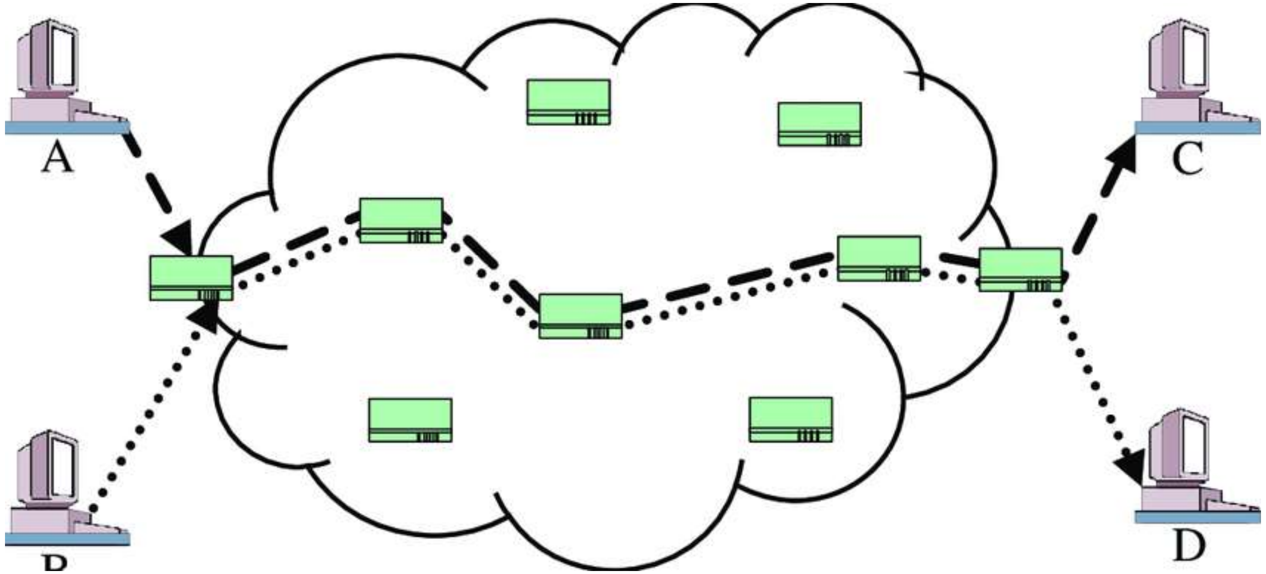


Circuit switching

- Circuit switching is a switching technique that establishes a dedicated path between sender and receiver.
- In the Circuit Switching Technique, once the connection is established then the dedicated path will remain to exist until the connection is terminated.
- A complete end-to-end path must exist before the communication takes place.
- After receiving the acknowledgment, dedicated path transfers the data, voice, video, a request signal from any user.
- Circuit switching is used in public telephone network. It is used for voice transmission.
- Fixed data can be transferred at a time in circuit switching technology.

Ckt Switching...

- Communication through circuit switching has 3 phases:
  - Circuit establishment
  - Data transfer
  - Circuit Disconnect

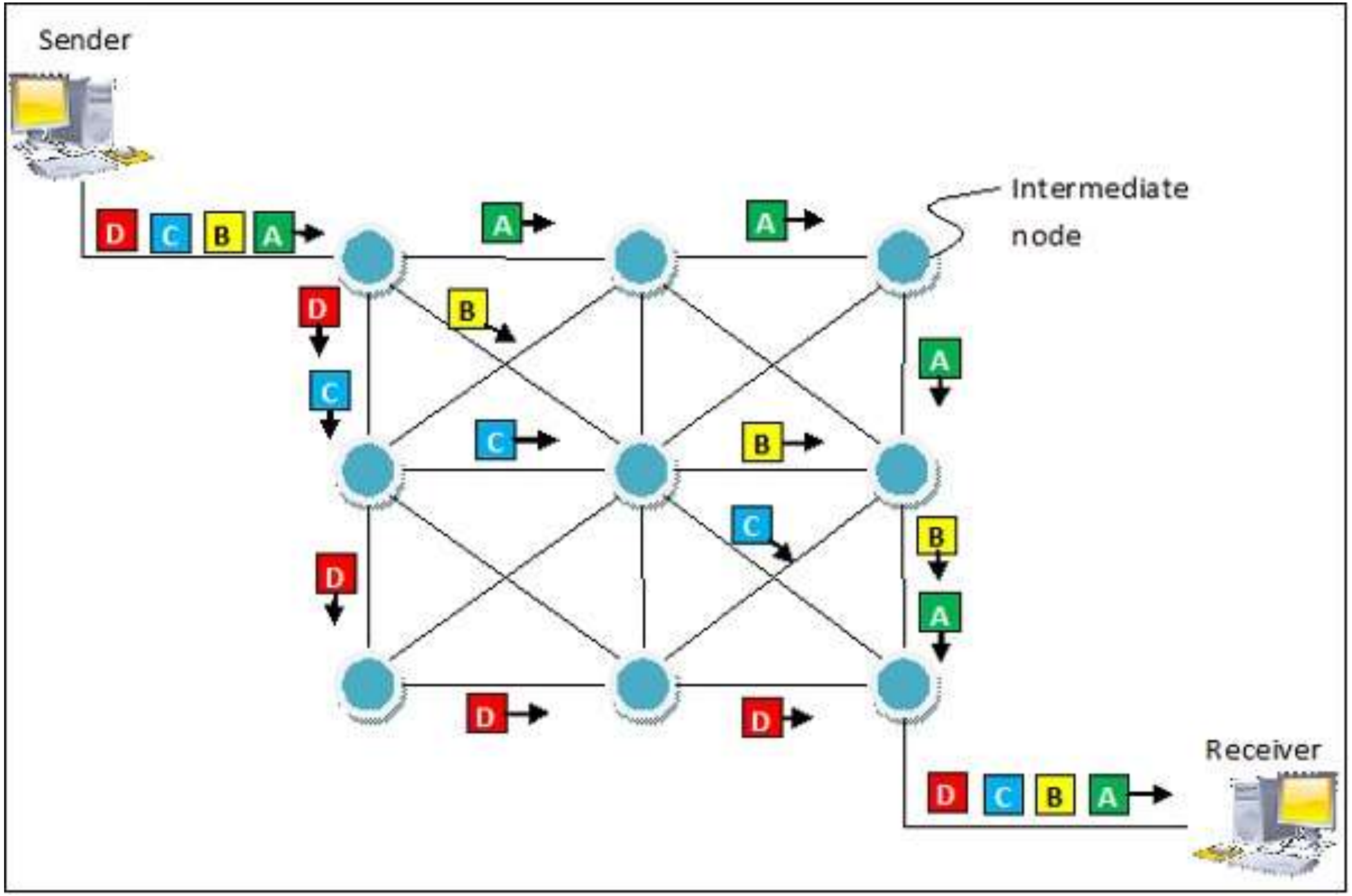


Ckt Switching…

- **Advantages**
  - Uses connection oriented services
  - Dedicated path/circuit provides guaranteed data delivery
  - Is suitable for long continuous transmission
- **Disadvantages**
  - Waste of bandwidth if its idle
  - Require more bandwidth
  - Time required to establish a link is too long (approx 10sec)
  - It is more expensive than other switching techniques as a dedicated path is required for each connection.

Packet switching

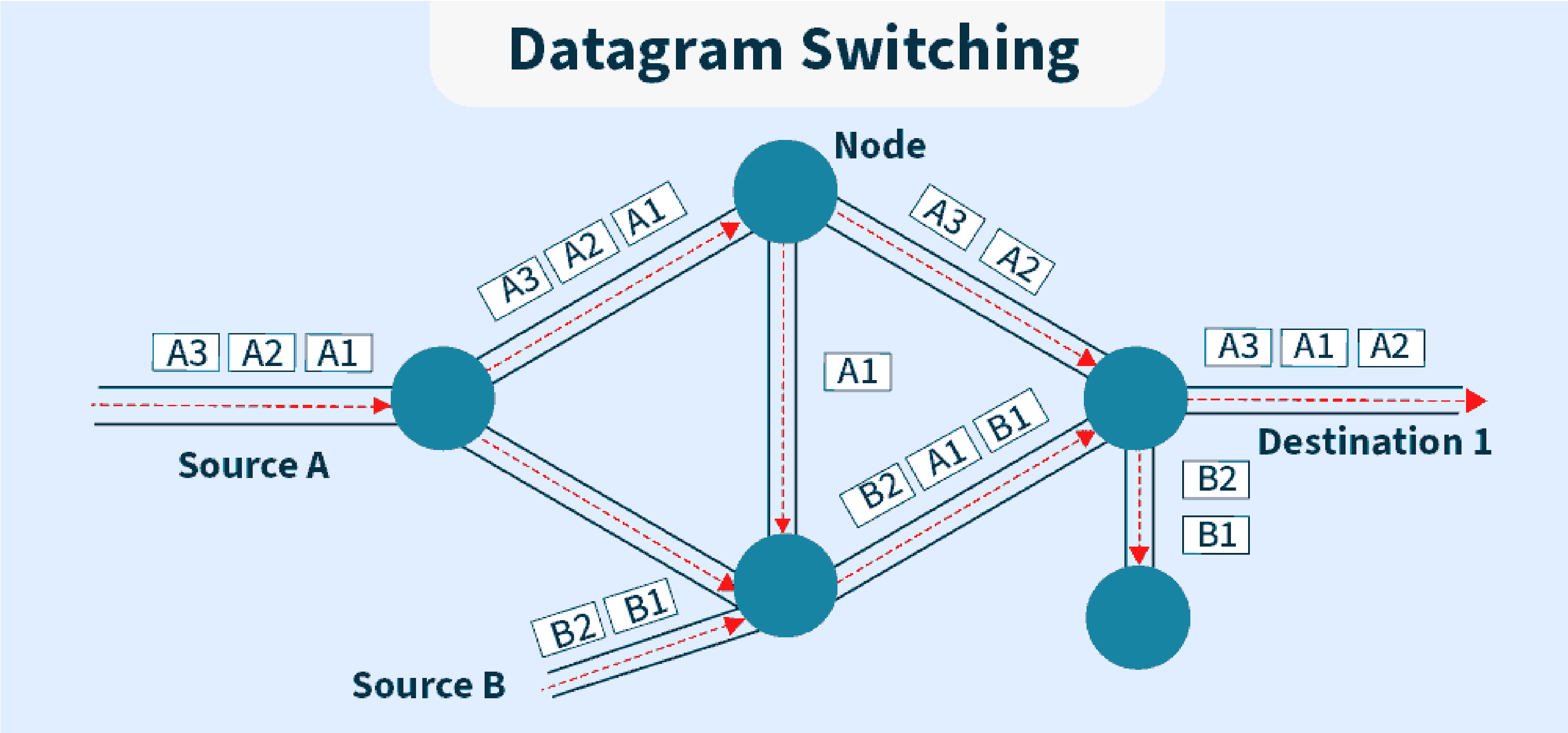
- Packet switching is a method of transferring the data to a network in form of packets.
- In order to transfer the file fast and efficient manner over the network and minimize the transmission latency, the data is broken into small pieces of variable length, called **Packet**.
- At the destination, all these small-parts (packets) has to be reassembled, belonging to the same file.


Sender
Intermediate node
Receiver
D C B A
A
A
D
B
A
C
C
B
C
B
D
A
D
D
D C B A

Datagram switching

- Datagram switching is a packet switching method that treats each packet, or datagram, as a separate entity.
- Each packet is routed via the network on its own. It is a service that does not require a connection.
- Because there is no specific channel for a connection session, there is no need to reserve resources.
- As a result, packets have a header with all the destination's information.
- The intermediate nodes assess a packet's header and select an appropriate link to a different node closer to the destination.

Datagram switching



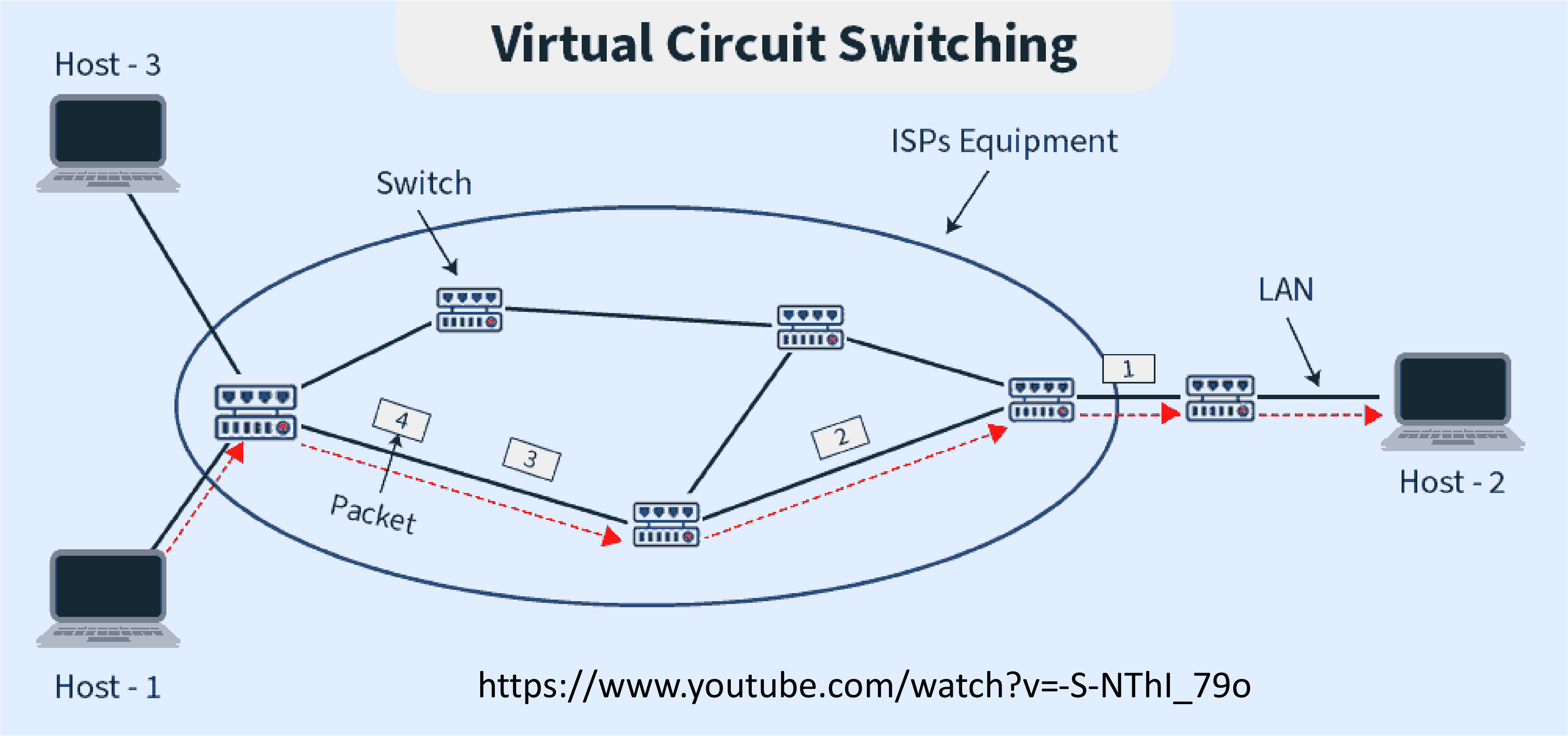
Advantages of Datagram Switching

- Scalability: Datagram switching is highly scalable and can handle large amounts of traffic on a network.
- Flexibility: Datagram switching is flexible and can support variable packet sizes and data rates.
- Simple routing: Datagram switching does not require a pre-established path for each packet, allowing packets to be routed dynamically.
- Lower latency: Datagram switching typically has lower latency than virtual circuit switching, as packets are sent immediately without any delay for setup.

Virtual packet switching

- Virtual packet switching approach in which a path is built between the source and the final destination through which all packets are routed throughout a call is known as virtual circuit switching.
- Because the connection looks to the user to be an infatuated physical circuit, this path is referred to as a virtual circuit. Other communications, on the other hand, may be sharing parts of the same path.
- Before the data transmission can commence, the source and destination must agree on a virtual circuit path. For the decision, all intermediary nodes between the two places add a routing entry to their routing database.
- Additional parameters, like the utmost packet size, also are exchanged between the source and therefore the destination during call setup.
- The virtual circuit is cleared after the info transfer is completed.

VCS


Virtual Circuit Switching
Host - 3
ISPs Equipment
Switch
LAN
1
4
3
2
Packet
Host - 2
Host - 1
https://www.youtube.com/watch?v=-S-NThI_79o

Advantage of Packet Switching over Circuit Switching

- High data transmission
- Minimal transmission latency
  - No time lost for establishing link
- Faster
- More efficient in terms of bandwidth, since the concept of reserving circuit is not there.
- More reliable as destination can detect the missing packet.
- More fault tolerant because packets may follow different path in case any link is down, Unlike Circuit Switching.
- Cost effective and comparatively cheaper to implement.

Disadvantage of Packet Switching over Circuit Switching

- Wrong order of packets
  - Packet Switching don't give packets in order, whereas Circuit Switching provides ordered delivery of packets because all the packets follow the same path.
- Packets may lost their routes
- Since the packets are unordered, we need to provide sequence numbers to each packet.
- Complexity is more at each node because of the facility to follow multiple path.
- Transmission delay is more because of rerouting.
- Packet Switching is beneficial only for small messages, but for bursty data (large messages) Circuit Switching is better.

Circuit Switching Vs Packet Switching

Circuit Switching	Packet Switching
Physical path between source and destination	No physical path
All packets use same path	Packets travel independently
Reserve the entire bandwidth in advance	Does not reserve
Bandwidth Wastage	No Bandwidth wastage
No store and forward transmission	Supports store and forward transmission

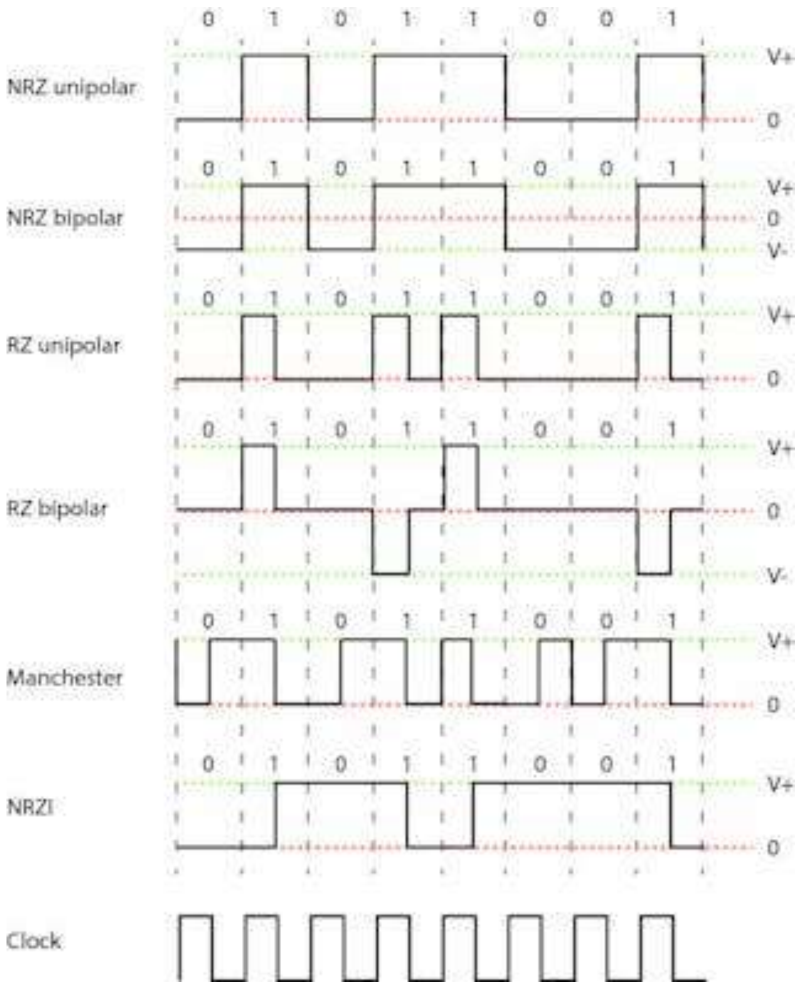
Circuit Switching	Packet Switching
Connection Oriented	Connection Less
Entire Message Have to follow same route during transmission	Entire Message can be divided and routed Independently
Implemented at Physical Layer	Implemented at Network Layer
Waste of bandwidth if Idle	No Waste of bandwidth if Idle
Initially designed for Voice Transmission	Initially designed for Data Transmission

3

*Ckt switching & Packet switching$_{(ch-2)}$

- Packet-switched networks enable end stations to dynamically share the network medium and the available bandwidth.
- Traditional phone networks, known as Public Switched Telephone Networks (PSTN) used circuit-switching.
  - In Circuit Switching, resources are reserved along the entire communication channel for the duration of the call. https://www.youtube.com/watch?v=B1tElYnFqL8 ckt vs pkt switching
- Conversely, Internet Protocol (IP) uses packet-switching.
  - In Packet Switching, information is digitally transmitted into one or more packets.
  - Packets know their destination, and may arrive there via different paths.

2.6 Data encoding: Manchester, NRZ-I, MLT3, 4B/5B



Manchester Encoding

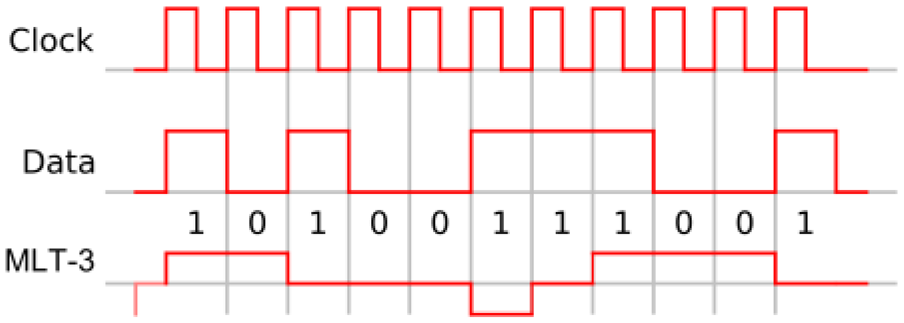
- How it works: A transition always occurs in the middle of each bit period (e.g., low-to-high for '0', high-to-low for '1'), ensuring clock synchronization (self-clocking).
- Pros: Excellent for clock recovery, no DC component.
- Cons: Requires double the bandwidth of NRZ.

NRZ-I (Non-Return-to-Zero Inverted)

- How it works: A '1' causes a voltage inversion (transition), while a '0' maintains the current voltage level (no transition).
- Pros: Lower bandwidth than Manchester.
- Cons: Long strings of '0's cause DC component issues and loss of clock sync.

MLT-3 (Multi-Level Transmit - 3 levels)

- How it works: Uses three voltage levels (+V, 0, -V). A '1' causes a transition to the next level in a cycle (+V -> 0 -> -V -> 0 -> +V...), while a '0' holds the current level.
- Pros: Significantly reduces bandwidth (1/4th of bit rate), reducing EMI, used in 100BASE-TX (Fast Ethernet).
- Cons: Can struggle with long strings of '0's for synchronization, needs other methods (like 4B/5B).



- MLT-3 cycles sequentially through the voltage levels −1, 0, +1, 0. It moves to the next state to transmit a 1 bit, and stays in the same state to transmit a 0 bit.
- Similar to simple NRZ encoding, MLT-3 has a coding efficiency of 1 bit/baud, however it requires four transitions (baud) to complete a full cycle (from low-to-middle, middle-to-high, high-to-middle, middle-to-low).
- Thus, the maximum fundamental frequency is reduced to one fourth of the baud rate. This makes signal transmission more amenable to copper wires.

4B/5B Encoding

- How it works: A block code that takes 4 bits of data and maps them to a unique 5-bit sequence, guaranteeing at least one transition in the code, preventing long strings of '0's.
- Pros: Solves DC balance and synchronization issues for underlying codes like NRZI, increases data rate overhead (80% efficiency).
- Cons: Adds overhead (5 bits for every 4 bits of data).
- 4B/5B: Converts 4-bit data nibbles into 5-bit symbols, adding transitions.
- MLT-3: Encodes these 5-bit symbols into a three-level signal to further reduce bandwidth.

- 4B/5B encoding is a line coding technique that maps every 4 bits of data (16 possible combinations) to a unique 5-bit code (out of 32 possibilities), achieving 80% efficiency, to prevent long strings of zeros, ensuring easier clock synchronization for receivers in networks like Fast Ethernet (100BASE-TX/FX) and FDDI, using specific 5-bit patterns for data and control signals like idle/start.
- How it Works
- Data Grouping: Input data is split into 4-bit chunks (nibbles).
- Mapping: Each 4-bit nibble is replaced by a specific 5-bit code from a predefined table.
- Key Rule: The 5-bit codes are chosen so that no sequence contains more than three consecutive zeros, preventing DC imbalance and aiding clock recovery.

Table 2.2.: 4B/5B encoding.

4-bit Data Symbol	5-bit Code
0000	11110
0001	01001
0010	10100
0011	10101
0100	01010
0101	01011
0110	01110
0111	01111
1000	10010
1001	10011
1010	10110
1011	10111
1100	11010
1101	11011
1110	11100
1111	11101